

5.2. TREATMENT: DEBRIDEMENT AND RETENTION OF IMPLANT

Authors: Marjan Wouthuyzen-Bakker, Ayman Ebied, Choe Hyonmin, Noam Shohat, Marei, Sameh

QUESTION 1: What are the indications and contraindications of using debridement, antibiotics and implant retention (DAIR) with exchange of modular components for the management of periprosthetic joint infection (PJI)?

RECOMMENDATION: The best advantage in performing DAIR of the prosthesis is seen in early postoperative PJI and acute hematogenous PJI, defined as symptoms existing for no longer than four weeks, and if the implant is stable. The KLIC and CRIME80 scores may aid in risk stratification.

LEVEL OF EVIDENCE: Moderate

DELEGATE VOTE: Agree: 80%, Disagree: 18%, Abstain: 2% (Super Majority, Strong Consensus)

RATIONALE

Open DAIR of the prosthesis is considered a less disruptive intervention that seeks to preserve a functional implant and forego the significant morbidity of implant removal and subsequent surgical procedures. While DAIR remains a viable and a less morbid alternative to resection arthroplasty, recent studies have demonstrated that an unsuccessful procedure is strongly associated with failure of future two-stage revision [1].

Strictly speaking, there are no absolute contraindications to perform a DAIR procedure, but a DAIR should be discouraged when the chance of failure without removing the implant is very high. Therefore, chronic PJIs should be considered an absolute contraindication to perform a DAIR procedure, as a fully developed mature biofilm with the presence of “persister cells” excludes the possibility for cure without removal of the implant [2,3]. Indeed, Barberan et al. demonstrated in 60 elderly patients with a Staphylococcal infection, that when the duration of symptoms exceeds one month, the failure rate increases exponentially when a conservative treatment is chosen without removal of the implant [4]. Although the efficacy of DAIR in chronic infections have been reported to be around 50% in a recent systematic review with a limited number of 29 patients, the average follow-up of these patients was only one year [5]. Extending the duration of antibiotic treatment following debridement does not seem to increase the chance for cure. Byren et al. clearly demonstrated that

prolonging antibiotic treatment for more than six months simply postpones, rather than prevents, failure [6]. For this reason, when the intention is to cure the PJI and the patient is medically fit for major surgery, chronic infections should undergo revision surgery with removal of hardware.

Failure rates following DAIR for acute PJI vary widely and range from 20 - 70%, with higher failure seen in acute hematogenous (late acute) PJI. Contraindications to performing a DAIR procedure in acute PJI are controversial. In general, all acute PJIs are candidates for debridement if the implant is well fixed, but several factors have been associated with an increased chance for failure. These factors include host and implant related factors, the severity and extensiveness of the infection, the duration of symptoms, the possibility to exchange the modular components during debridement and the causative microorganism [1,7–40]. In order to avoid surgery that has a very high risk of failure, selecting a subset of patients that are more likely to benefit from revision surgery instead of DAIR, would be helpful. A preoperative risk score has been developed to predict failure following DAIR for early acute Kidney, Liver, Index surgery, Cemented prosthesis and C-reactive protein value (KLIC-score, Fig. 1A) and acute hematogenous PJIs (CRIME80 score, Fig. 1B) [27,30]. These preoperative scoring systems could be used in clinical practice to select those patients who are most eligible for DAIR.

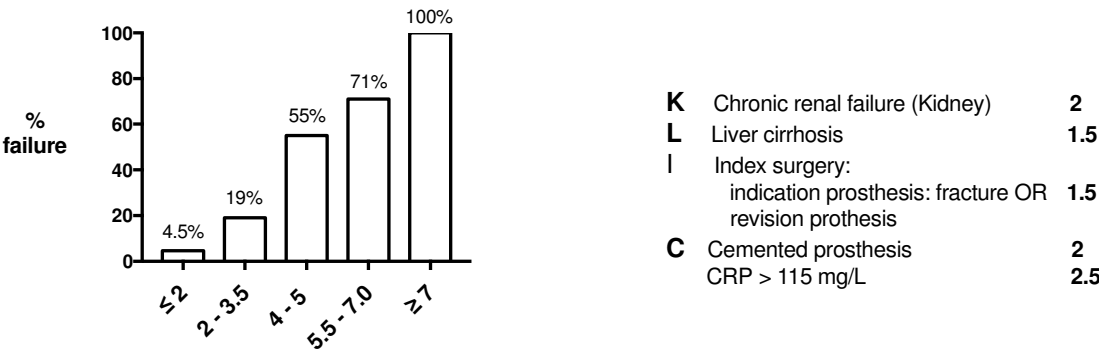


FIGURE 1A. KLIC preoperative risk score [27,30]

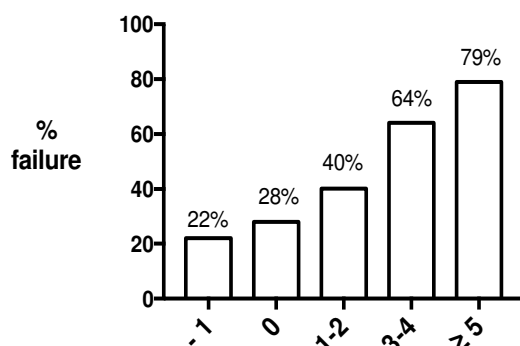


FIGURE 1B. CRIME80 preoperative risk score [27,30]

C	COPD	2
	CRP > 150 mg/L	1
R	Rheumatoid arthritis	3
I	Indication prosthesis: fracture	3
M	Male	1
E	Exchange of mobile components	-1
80	Age > 80 years	2

REFERENCES

- Tsang STJ, Ting J, Simpson AHRW, Gaston P. Outcomes following debridement, antibiotics and implant retention in the management of periprosthetic infections of the hip: a review of cohort studies. *Bone Joint J.* 2017;99-B:1458-1466. doi:10.1302/0301-620X.99B11.BJl-2017-0088.R1.
- Lebeaux D, Ghigo JM, Beloin C. Biofilm-related infections: bridging the gap between clinical management and fundamental aspects of recalcitrance toward antibiotics. *Microbiol Mol Biol Rev.* 2014;78:510-543. doi:10.1128/MMBR.00013-14.
- Davies D. Understanding biofilm resistance to antibacterial agents. *Nat Rev Drug Discov.* 2003;2:114-122. doi:10.1038/nrd1008.
- Barberán J, Aguilar L, Carroquino G, Giménez M-J, Sánchez B, Martínez D, et al. Conservative treatment of staphylococcal prosthetic joint infections in elderly patients. *Am J Med.* 2006;119:993.e7-e10. doi:10.1016/j.amjmed.2006.03.036.
- Maillet M, Pavese P, Bruley D, Seigneurin A, François P. Is prosthesis retention effective for chronic infections in hip arthroplasties? A systematic literature review. *Eur J Clin Microbiol Infect Dis.* 2015;34:1495-1502. doi:10.1007/s10096-015-2388-8.
- Byren I, Bejon P, Atkins BL, Angus B, Masters S, McLardy-Smith P, et al. One hundred and twelve infected arthroplasties treated with "DAIR" (debridement, antibiotics and implant retention): antibiotic duration and outcome. *J Antimicrob Chemother.* 2009;63:1264-1271. doi:10.1093/jac/dkp107.
- Grammatopoulos G, Bolduc M-E, Atkins BL, Kendrick BJL, McLardy-Smith P, Murray DW, et al. Functional outcome of debridement, antibiotics and implant retention in periprosthetic joint infection involving the hip: a case-control study. *Bone Joint J.* 2017;99-B:614-622. doi:10.1302/0301-620X.99B5.BJl-2016-0562.R2.
- Zhang C, Yan CH, Chan PK, Ng FY, Chiu KY. Polyethylene insert exchange is crucial in debridement for acute periprosthetic infections following total knee arthroplasty. *J Knee Surg.* 2017;30:36-41. doi:10.1055/s-0036-1579667.
- Choi HR, von Knoch F, Zurakowski D, Nelson SB, Malchau H. Can implant retention be recommended for treatment of infected TKA? *Clin Orthop Relat Res.* 2011;469:961-969. doi:10.1007/s11999-010-1679-8.
- Lora-Tamayo J, Murillo O, Iribarren JA, Soriano A, Sánchez-Somolinos M, Baraia-Etxaburu JM, et al. A large multicenter study of methicillin-susceptible and methicillin-resistant *Staphylococcus aureus* prosthetic joint infections managed with implant retention. *Clin Infect Dis.* 2013;56:182-194. doi:10.1093/cid/cis746.
- Lora-Tamayo J, Senneville É, Ribera A, Bernard L, Dupon M, Zeller V, et al. The not-so-good prognosis of streptococcal periprosthetic joint infection managed by implant retention: the results of a large multicenter study. *Clin Infect Dis.* 2017;64:1742-1752. doi:10.1093/cid/cix227.
- Rodríguez-Pardo D, Pigrau C, Lora-Tamayo J, Soriano A, del Toro MD, Cobo J, et al. Gram-negative prosthetic joint infection: outcome of a debridement, antibiotics and implant retention approach. A large multicentre study. *Clin Microbiol Infect.* 2014;20:O911-O919. doi:10.1111/1469-0691.12649.
- Urish KL, Bullock AG, Kreger AM, Shah NB, Jeong K, Rothenberger SD, et al. A multicenter study of irrigation and debridement in total knee arthroplasty periprosthetic joint infection: treatment failure is high. *J Arthroplasty.* 2018;33:1154-1159. doi:10.1016/j.arth.2017.11.029.
- Koh IJ, Han SB, In Y, Oh KJ, Lee DH, Kim TK, et al. Open debridement and prosthesis retention is a viable treatment option for acute periprosthetic joint infection after total knee arthroplasty. *Arch Orthop Trauma Surg.* 2015;135:847-855. doi:10.1007/s00402-015-2237-3.
- Triantafyllopoulos GK, Poultides LA, Sakellariou VI, Zhang W, Scolio PK, Ma Y, et al. Irrigation and debridement for periprosthetic infections of the hip and factors determining outcome. *Int Orthop.* 2015;39:1203-1209. doi:10.1007/s00264-015-2753-3.
- Kuiper JWP, Vos SJC, Saouti R, Vergroesen DA, Graat HCA, Debets-Ossenkopp YJ, et al. Prosthetic joint-associated infections treated with DAIR (debridement, antibiotics, irrigation, and retention): analysis of risk factors and local antibiotic carriers in 91 patients. *Acta Orthop.* 2013;84:380-386. doi:10.3109/17453674.2013.823589.
- Marculescu CE, Berbari EF, Hanssen AD, Steckelberg JM, Harmsen SW, Mandrekar JN, et al. Outcome of prosthetic joint infections treated with debridement and retention of components. *Clin Infect Dis.* 2006;42:471-478. doi:10.1086/499234.
- Buller LT, Sabry FY, Easton RW, Klika AK, Barsoum WK. The preoperative prediction of success following irrigation and debridement with polyethylene exchange for hip and knee prosthetic joint infections. *J Arthroplasty.* 2012;27:857-864.e1-e4. doi:10.1016/j.arth.2012.01.003.
- Hsieh PH, Lee MS, Hsu KY, Chang -H, Shih HN, Ueng SW. Gram-negative prosthetic joint infections: risk factors and outcome of treatment. *Clin Infect Dis.* 2009;49:1036-1043. doi:10.1086/605593.
- Tornero E, Morata L, Martínez-Pastor JC, Angulo S, Combalia A, Bori G, et al. Importance of selection and duration of antibiotic regimen in prosthetic joint infections treated with debridement and implant retention. *J Antimicrob Chemother.* 2016;71:1395-1401. doi:10.1093/jac/dkv481.
- Puhto A-P, Puhto T, Niinimäki T, Ohtonen P, Leppilähti J, Syrjälä H. Predictors of treatment outcome in prosthetic joint infections treated with prosthesis retention. *Int Orthop.* 2015;39:1785-1791. doi:10.1007/s00264-015-2819-2.
- Holmberg A, Thórhallsdóttir VG, Robertsson O, W-Dahl A, Stefánsdóttir A. 75% success rate after open debridement, exchange of tibial insert, and antibiotics in knee prosthetic joint infections. *Acta Orthop.* 2015;86:457-462. doi:10.3109/17453674.2015.1026756.
- Vilchez F, Martínez-Pastor JC, García-Ramiro S, Bori G, Tornero E, García E, et al. Efficacy of debridement in hematogenous and early post-surgical prosthetic joint infections. *Int J Artif Organs.* 2011;34:863-869. doi:10.5301/ijao.5000029.
- El Helou OC, Berbari EF, Lahr BD, Eckel-Passow JE, Razonable RR, Sia IG, et al. Efficacy and safety of rifampin containing regimen for staphylococcal prosthetic joint infections treated with debridement and retention. *Eur J Clin Microbiol Infect Dis.* 2010;29:961-967. doi:10.1007/s10096-010-0952-9.
- Martínez-Pastor JC, Muñoz-Mahamud E, Vilchez F, García-Ramiro S, Bori G, Sierra J, et al. Outcome of acute prosthetic joint infections due to gram-negative bacilli treated with open debridement and retention of the prosthesis. *Antimicrob Agents Chemother.* 2009;53:4772-4777. doi:10.1128/AAC.00188-09.
- Zimmerli W, Widmer AF, Blatter M, Frei R, Ochsen PE. Role of rifampin for treatment of orthopedic implant-related staphylococcal infections: a randomized controlled trial. *Foreign-Body Infection (FBI) Study Group. JAMA.* 1998;279:1537-1541.
- Tornero E, Morata L, Martínez-Pastor JC, Bori G, Climent C, García-Velez DM, et al. KLIC-score for predicting early failure in prosthetic joint infections treated with debridement, implant retention and antibiotics. *Clin Microbiol Infect.* 2015;21:786.e9-786.e17. doi:10.1016/j.cmi.2015.04.012.
- Senneville E, Joulie D, Legout L, Valette M, Dezèque H, Beltrand E, et al. Outcome and predictors of treatment failure in total hip/knee prosthetic joint infections due to *Staphylococcus aureus*. *Clin Infect Dis.* 2011;53:334-340. doi:10.1093/cid/cir402.
- Grossi O, Asseray N, Bourigault C, Corvec S, Valette M, Navas D, et al. Gram-negative prosthetic joint infections managed according to a multidisciplinary standardized approach: risk factors for failure and outcome with and without fluoroquinolones. *J Antimicrob Chemother.* 2016;71:2593-2597. doi:10.1093/jac/dkw202.

- [30] Löwik CAM, Jutte PC, Tornero E, Ploegmakers JJW, Knobben BAS, de Vries AJ, et al. Predicting failure in early acute prosthetic joint infection treated with debridement, antibiotics, and implant retention: external validation of the KLIC Score. *J Arthroplasty*. 2018. doi:10.1016/j.arth.2018.03.041.
- [31] Hsieh PH, Huang KC, Shih HN. Prosthetic joint infection in patients with rheumatoid arthritis: an outcome analysis compared with controls. *PLoS ONE*. 2013;8:e71666. doi:10.1371/journal.pone.0071666.
- [32] Son WS, Shon OJ, Lee DC, Park SJ, Yang HS. Efficacy of open debridement and polyethylene exchange in strictly selected patients with infection after total knee arthroplasty. *Knee Surg Relat Res*. 2017;29:172–179. doi:10.5792/ksrr.16.040.
- [33] Tornero E, Martínez-Pastor JC, Bori G, García-Ramiro S, Morata L, Bosch J, et al. Risk factors for failure in early prosthetic joint infection treated with debridement. Influence of etiology and antibiotic treatment. *J Appl Biomater Funct Mater*. 2014;12:129–134. doi:10.5301/jabfm.5000209.
- [34] Bergkvist M, Mukka SS, Johansson L, Ahl TE, Sayed-Noor AS, Sköldenberg OG, et al. Debridement, antibiotics and implant retention in early periprosthetic joint infection. *Hip Int*. 2016;26:138–143. doi:10.5301/hipint.5000328.
- [35] Vilchez F, Martínez-Pastor JC, García-Ramiro S, Bori G, Maculé F, Sierra J, et al. Outcome and predictors of treatment failure in early post-surgical prosthetic joint infections due to *Staphylococcus aureus* treated with debridement. *Clin Microbiol Infect*. 2011;17:439–444. doi:10.1111/j.1469-0691.2010.03244.x.
- [36] Rodríguez D, Pigrau C, Euba G, Cobo J, García-Lechuz J, Palomino J, et al. Acute hematogenous prosthetic joint infection: prospective evaluation of medical and surgical management. *Clin Microbiol Infect*. 2010;16:1789–1795. doi:10.1111/j.1469-0691.2010.03157.x.
- [37] Cobo J, Miguel LGS, Euba G, Rodríguez D, García-Lechuz JM, Riera M, et al. Early prosthetic joint infection: outcomes with debridement and implant retention followed by antibiotic therapy. *Clin Microbiol Infect*. 2011;17:1632–1637. doi:10.1111/j.1469-0691.2010.03333.x.
- [38] Tande AJ, Palraj BR, Osmon DR, Berbari EF, Baddour LM, Lohse CM, et al. Clinical presentation, risk factors, and outcomes of hematogenous prosthetic joint infection in patients with *Staphylococcus aureus* Bacteremia. *Am J Med*. 2016;129:221.e11–e20. doi:10.1016/j.amjmed.2015.09.006.
- [39] Letouvet B, Arvieux C, Leroy H, Polard JL, Chaplain JM, Common H, et al. Predictors of failure for prosthetic joint infections treated with debridement. *Med Mal Infect*. 2016;46:39–43. doi:10.1016/j.medmal.2015.11.007.
- [40] Soriano A, García S, Bori G, Almela M, Gallart X, Maculé F, et al. Treatment of acute post-surgical infection of joint arthroplasty. *Clin Microbiol Infect*. 2006;12:930–933. doi:10.1111/j.1469-0691.2006.01463.x.



Authors: Anna Stefánsdóttir, Georgios Komnos

QUESTION 2: Is debridement, antibiotics and implant retention (DAIR) an emergency procedure for patients with acute periprosthetic joint infection (PJI) or should patient optimization be implemented prior to surgery to enhance the success of this procedure?

RECOMMENDATION: DAIR is not an emergency procedure but should be performed on an urgent basis when the patient with acute PJI is medically and surgically optimized.

LEVEL OF EVIDENCE: Limited

DELEGATE VOTE: Agree: 97%, Disagree: 3%, Abstain: 0% (Unanimous, Strongest Consensus)

RATIONALE

At the present time DAIR is reserved for patients with acute PJIs when no loosening of the implants is present [1,2]. Success rates vary among different studies from 16%–82% [3–7]. The large majority of studies regarding DAIR focus on reporting the success rates or evaluating the factors that are correlated with success [2,4,6,8–16]. However, none of these studies have focused on the urgency of DAIR as a procedure.

DAIR should be considered an urgent, but not emergent procedure, as the time period from the onset of symptoms until the operation has been reported to be important factor affecting the success of the procedure [5]. Factors that are known to affect the outcome of DAIR include the type of infecting organism [5,10,17–21], duration of symptoms before intervention [4,7,11–13,17,20,21], type and duration of antibiotic therapy [6,14,22], age [11], erythrocyte sedimentation rate (ESR) values at presentation [4,13,19,20], presence of underlying inflammatory conditions [4,19], exchange of modular components [7,17,23] and the presence of preoperative comorbidities like anemia [24].

An exact cutoff time beyond which DAIR should not be attempted has not been determined. Nevertheless, the duration of symptoms less than one week has been correlated to a higher success rate [4,5,7,12,17,21]. Furthermore, age of implant ≤ 15 days has been identified as a prognostic factor for successful DAIR [25].

There are patient-related factors and medical comorbidities, which, if not controlled, may result in severe complications and failure of the procedure. Comorbidities, such as rheumatoid arthritis, are not possible to adjust prior to debridement. However, correction of malnutrition, coagulopathy, anemia, hyperglycemia and diabetes should be pursued. Subjecting a patient to irrigation

and debridement (I&D) without addressing an underlying coagulopathy could result in the development of a subsequent hematoma and its adverse effects. Thus, it is critical that conditions such as coagulopathy, nutritional status, uncontrolled hyperglycemia (>200 mg/ml), severe anemia (hemoglobin <10 mg/dL) and other reversible conditions are addressed prior to subjecting a patient to DAIR.

In conclusion, we therefore recommend that patients with acute PJI are evaluated on an urgent basis and the surgery is performed when patient is optimized from medical and surgical perspective.

REFERENCES

- [1] Osmon DR, Berbari EF, Berendt AR, Lew D, Zimmerli W, Steckelberg JM, et al. Diagnosis and management of prosthetic joint infection: Clinical practice guidelines by the infectious diseases Society of America. *Clin Infect Dis*. 2013;56. doi:10.1093/cid/cis803.
- [2] Achermann Y, Stasch P, Preiss S, Lucke K, Vogt M. Characteristics and treatment outcomes of 69 cases with early prosthetic joint infections of the hip and knee. *Infection*. 2014;42:511–519. doi:10.1007/s15010-014-0584-6.
- [3] Bradbury T, Fehring TK, Taunton M, Hanssen A, Azzam K, Parvizi J, et al. The fate of acute methicillin-resistant *Staphylococcus aureus* periprosthetic knee infections treated by open debridement and retention of components. *J Arthroplasty*. 2009;24:101–104. doi:10.1016/j.arth.2009.04.028.
- [4] Kuiper JWP, Vos SJ, Saouti R, Vergroesen DA, Graat HCA, Debets-Ossenkopp YJ, et al. Prosthetic joint-associated infections treated with DAIR (debridement, antibiotics, irrigation, and retention). *Acta Orthop*. 2013;84:380–386. doi:10.3109/17453674.2013.823589.
- [5] Urish KL, Bullock AG, Kreger AM, Shah NB, Jeong K, Rothenberger SD, et al. A multicenter study of irrigation and debridement in total knee arthroplasty periprosthetic joint infection: treatment failure is high. *J Arthroplasty*. 2017. doi:10.1016/j.arth.2017.11.029.
- [6] Byren I, Bejon P, Atkins BL, Angus B, Masters S, McLardy-Smith P, et al. One hundred and twelve infected arthroplasties treated with “DAIR” (debridement, antibiotics and implant retention): Antibiotic duration and outcome. *J Antimicrob Chemother*. 2009;63:1264–1271. doi:10.1093/jac/dkp107.

- [7] Tsang STJ, Ting J, Simpson AHRW, Gaston P. Outcomes following debridement, antibiotics and implant retention in the management of periprosthetic infections of the hip: a review of cohort studies. *Bone Joint J*. 2017;99B:1458–1466. doi:10.1302/0301-620X.99B11.BJJ-2017-0088.R1.
- [8] Cobo J, Miguel LGS, Euba G, Rodríguez D, García-Lechuz JM, Riera M, et al. Early prosthetic joint infection: outcomes with debridement and implant retention followed by antibiotic therapy. *Clin Microbiol Infect*. 2011;17:1632–1637. doi:10.1111/j.1469-0691.2010.03333.x.
- [9] Matthews PC, Berendt AR, McNally MA, Byren I. Diagnosis and management of prosthetic joint infection. *BMJ*. 2009;338:b1773. doi:10.1136/bmj.b1773.
- [10] Duque AF, Post ZD, Lutz RW, Orozco FR, Pulido SH, Ong AC. Is there still a role for irrigation and debridement with liner exchange in acute periprosthetic total knee infection? *J Arthroplasty*. 2017;32:1280–1284. doi:10.1016/j.arth.2016.10.029.
- [11] de Vries L, van der Weegen W, Neve W, Das H, Ridwan B, Steens J. The effectiveness of debridement, antibiotics and irrigation for periprosthetic joint infections after primary hip and knee arthroplasty. A 15 years retrospective study in two community hospitals in the Netherlands. *J Bone Jt Infect*. 2016;1:20–24.
- [12] Koh IJ, Han SB, In Y, Oh KJ, Lee DH, Kim TK. Open debridement and prosthesis retention is a viable treatment option for acute periprosthetic joint infection after total knee arthroplasty. *Arch Orthop Trauma Surg*. 2015;135:847–855. doi:10.1007/s00402-015-2237-3.
- [13] Klare CM, Fortney TA, Kahng PW, Cox AP, Keeney BJ, Moschetti WE. Prognostic factors for success after irrigation and debridement with modular component exchange for infected total knee arthroplasty. *J Arthroplasty*. 2018. doi:10.1016/j.arth.2018.02.004.
- [14] Puhto AP, Puhto T, Niinimäki T, Ohtonen P, Leppilähti J, Syrjälä H. Predictors of treatment outcome in prosthetic joint infections treated with prosthesis retention. *Int Orthop*. 2015;39:1785–1791. doi:10.1007/s00264-015-2819-2.
- [15] Sendi P, Löttscher PO, Kessler B, Graber P, Zimmerli W, Clauss M. Debridement and implant retention in the management of hip periprosthetic joint infection. *Bone Joint J*. 2017;99B:330–336. doi:10.1302/0301-620X.99B3.BJJ-2016-0609.R1.
- [16] Anagnostakos K. Can periprosthetic hip joint infections be successfully managed by debridement and prosthesis retention? *World J Orthop*. 2014;5:218. doi:10.5312/wjo.v5.i3.218.
- [17] Qasim SN, Swann A, Ashford R. The DAIR (debridement, antibiotics and implant retention) procedure for infected total knee replacement – a literature review. *SICOT-J*. 2017;3:2. doi:10.1051/sicotj/2016038.
- [18] Choi HR, Von Knoch F, Kandil AO, Zurakowski D, Moore S, Malchau H. Retention treatment after periprosthetic total hip arthroplasty infection. *Int Orthop*. 2012;36:723–729. doi:10.1007/s00264-011-1324-5.
- [19] Kuiper JW. Treatment of acute periprosthetic infections with prosthesis retention: Review of current concepts. *World J Orthop*. 2014;5:667. doi:10.5312/wjo.v5.i5.667.
- [20] Buller IT, Sabry FY, Easton RW, Klika AK, Barsoum WK. The preoperative prediction of success following irrigation and debridement with polyethylene exchange for hip and knee prosthetic joint infections. *J Arthroplasty*. 2012;27. doi:10.1016/j.arth.2012.01.003.
- [21] Marculescu CE, Berbari EF, Hanssen AD, Steckelberg JM, Harmsen SW, Mandrekar JN, et al. Outcome of prosthetic joint infections treated with debridement and retention of components. *Clin Infect Dis*. 2006;42:471–478. doi:10.1086/499234.
- [22] Chaussade H, Uçkay I, Vuagnat A, Druon J, Gras G, Rosset P, et al. Antibiotic therapy duration for prosthetic joint infections treated by debridement and implant retention (DAIR): similar long-term remission for 6 weeks as compared to 12 weeks. *Int J Infect Dis*. 2017;63:37–42. doi:10.1016/j.ijid.2017.08.002.
- [23] Lora-Tamayo J, Murillo O, Iribarren JA, Soriano A, Sánchez-Somolinos M, Baraia-Etxaburu JM, et al. A large multicenter study of methicillin-susceptible and methicillin-resistant *Staphylococcus aureus* prosthetic joint infections managed with implant retention. *Clin Infect Dis*. 2013;56:182–194. doi:10.1093/cid/cis746.
- [24] Swenson RD, Butterfield JA, Irwin TJ, Zurlo JJ, Davis CM. Preoperative anemia is associated with failure of open debridement polyethylene exchange in acute and acute hematogenous prosthetic joint infection. *J Arthroplasty*. 2018. doi:10.1016/j.arth.2018.01.042.
- [25] Tornero E, Martínez-Pastor JC, Bori G, García-Ramiro S, Morata L, Bosch J, et al. Risk factors for failure in early prosthetic joint infection treated with debridement. Influence of etiology and antibiotic treatment. *J Appl Biomater Funct Mater*. 2014;12:129–134. doi:10.5301/jabfm.5000209.



Authors: Jaime Lora-Tamayo, Benjamin Zmistowski, Mikel Mancheno-Losa

QUESTION 3: Does identification of the pathogen prior to performing debridement, antibiotics and implant retention (DAIR) help guide the surgeon's decision making? If so, should you wait, in a clinically stable patient, until the pathogen has been identified?

RECOMMENDATION: The identification of the responsible microorganism before DAIR is desirable. However, it should not prevent timely surgical intervention if delay in surgery is believed to promote further establishment of biofilm formation and compromise the outcome of surgical intervention.

LEVEL OF EVIDENCE: Limited

DELEGATE VOTE: Agree: 94%, Disagree: 4%, Abstain: 2% (Super Majority, Strong Consensus)

RATIONALE

In implant related infections, the need for use of targeted antibiotics with proven action against the infecting pathogen and penetration into the biofilm has been suggested [1]. For instance, experts would likely agree DAIR is appropriate when ciprofloxacin-susceptible *Escherichia coli* is the infecting organism but, would probably discourage DAIR if the infective organism is a *Candida* spp. Thus, from a general perspective, knowledge of the pathogen prior to surgical intervention is desired. However, the real debate is whether waiting to determine the infective organism would adversely affect the outcome of DAIR and the timely intervention. The answer to this question requires an understanding of the implications of delaying DAIR and the consequences of performing DAIR without knowledge of the infecting pathogen.

Regarding the issue of time, Infectious Diseases Society of America (IDSA) guidelines, in conjunction with other authors, recommend a maximum of 21 days of symptom duration before

utilizing DAIR to treat periprosthetic joint infection (PJI) [1,2]. This time limit, which has not been identified in comparative studies, is the same as that used in the pivotal clinical trial by Zimmerli et al. on the use of rifampin: none of the patients included in that cohort underwent DAIR beyond 21 days [3]. However, it remains uncertain whether these patients could have benefited from therapy if they had been submitted to DAIR more than 21 days after the beginning of symptoms. To this end, many observational studies have tried to find a precise cut-off of symptom duration, but heterogeneous populations with poorly reproduced results have emerged. Brand et al. observed that as little as a two-day delay in performing DAIR would significantly increase the odds of failure in a cohort of patients with staphylococcal PJI, mainly managed with β -lactams [4]. Other studies have also observed a poor outcome among patients with longer duration of symptoms without identifying a reliable time limit [5–13].

Inability to establish an optimal time threshold for DAIR may be mainly due to two causes. First, a short interval of time for performing DAIR may be a surrogate marker of severity of illness, since patients with sepsis or bacteremia are usually operated on sooner than more stable cases. Ill patients have a higher likelihood of failure [12,14], causing a short duration of symptoms to be paradoxically associated with a worse prognosis. Second, the duration of symptoms may be difficult to establish, especially in post-surgical cases where the postoperative inflammatory signs and pain may overlap the symptoms of infection. In these post-surgical cases, the prosthesis age before DAIR (i.e., the time from prosthesis placement to debridement) may be a more reliable variable. Yet, there is controversy on the definition of an early post-surgical infection that could be managed by DAIR. While IDSA guidelines do not recommend DAIR for patients with PJI that started greater than one month from the index arthroplasty [2], other important studies and the First International Consensus extend this period to three months [1,15]. Two large studies including staphylococcal and streptococcal PJI managed with DAIR found no differences in recurrent infection with a prosthesis age of less than one month versus those that were one to three months old [12,13]. Overall, it seems reasonable to assume that the sooner the DAIR is performed, the better the outcome will be, but there is insufficient evidence to recommend a specific time-limit of symptoms duration beyond which DAIR should be discouraged.

Bearing these considerations in mind, the question falls back onto the influence of the type of infecting microorganism(s) and its antibiotic susceptibility profile on prognosis. Apart from particular and rare situations such as the fungal infection previously mentioned or other multi-drug resistant bacteria, there is limited consensus on the impact of organism type on the outcomes of DAIR. Wide ranges of clinical success rates have been reported for common pathogens when managed by DAIR: 13%–90% for *Staphylococcus aureus* [4,6,14,16–18], 27%–94% for gram-negative bacilli (GNB) [8,14,17] and 40%–94% for streptococci [19–24]. The largest observational studies performed to date set these cure rates in 55% for *S. aureus* [12], 58% for streptococci [13], 51% for enterococci [25] and 68% for GNB (with significant differences between fluoroquinolone-susceptible and -resistant strains: 79% vs. 40%, respectively) [26].

Whether a 50% risk of failure should discourage use of DAIR is a matter of controversy. In older patients, Fisman et al. suggested an annual relapse rate $\approx$ 30% after DAIR to be cost-effective when compared with a two-step exchange procedure [27]. The potential advantages of a successful DAIR (one surgery, bone-stock preservation and less economic costs) [28] should be balanced with the consequences of failure. In this regard, conflicting results have been reported on the consequences of a failed DAIR. Sherrel et al. observed a higher likelihood of relapse among patients undergoing a two-stage revision after a non-successful DAIR, as compared with patients submitted to an elective two-stage exchange procedure [29]. However, these results have been contested by two other observational studies [30,31]. Furthermore, functional outcome has been reported to be identical in patients undergoing two-stage after failed DAIR compared to patients undergoing direct two-stage exchange [30, 31].

In summation, the type of infecting pathogen can be valuable information in the treatment algorithm for patients and surgeons considering DAIR. However, a prompt surgery is also of utmost importance. Therefore, the efforts to identify the causative pathogen for PJI should not cause undue delay in timely surgical intervention. Often, the pathogens of concern are virulent in nature and usually identified soon after culture samples are processed and cultured.

REFERENCES

- Zimmerli W, Trampuz A, Ochsner PE. Prosthetic-joint infections. *N Engl J Med*. 2004;351:1645–1654. doi:10.1056/NEJMra040181.
- Osmon DR, Berbari EF, Berendt AR, Lew D, Zimmerli W, Steckelberg JM, et al. Diagnosis and management of prosthetic joint infection: clinical practice guidelines by the Infectious Diseases Society of America. *Clin Infect Dis*. 2013;56:e1–e25. doi:10.1093/cid/cis803.
- Zimmerli W, Widmer AF, Blatter M, Frei R, Ochsner PE. Role of rifampin for treatment of orthopedic implant-related staphylococcal infections: a randomized controlled trial. *Foreign-Body Infection (FBI) Study Group*. *JAMA*. 1998;279:1537–1541.
- Brandt CM, Duffy MC, Berbari EF, Hanssen AD, Steckelberg JM, Osmon DR. Staphylococcus aureus prosthetic joint infection treated with prosthesis removal and delayed reimplantation arthroplasty. *Mayo Clin Proc*. 1999;74:553–558. doi:10.4065/74.6.553.
- Burger RR, Basch T, Hopson CN. Implant salvage in infected total knee arthroplasty. *Clin Orthop Relat Res*. 1991;105–112.
- Barberán J, Aguilar L, Carroquino G, Giménez M-J, Sánchez B, Martínez D, et al. Conservative treatment of staphylococcal prosthetic joint infections in elderly patients. *Am J Med*. 2006;119:993.e7–e10. doi:10.1016/j.amjmed.2006.03.036.
- Geurts JAP, Janssen DMC, Kessels AGH, Walenkamp GHM. Good results in postoperative and hematogenous deep infections of 89 stable total hip and knee replacements with retention of prosthesis and local antibiotics. *Acta Orthop*. 2013;84:509–516. doi:10.3109/17453674.2013.858288.
- Hsieh PH, Huang KC, Lee PC, Lee MS. Two-stage revision of infected hip arthroplasty using an antibiotic-loaded spacer: retrospective comparison between short-term and prolonged antibiotic therapy. *J Antimicrob Chemother*. 2009;64:392–397. doi:10.1093/jac/dkp177.
- Marculescu CE, Berbari EF, Hanssen AD, Steckelberg JM, Harmsen SW, Mandrekar JN, et al. Outcome of prosthetic joint infections treated with debridement and retention of components. *Clin Infect Dis*. 2006;42:471–478. doi:10.1086/499234.
- Schoifet SD, Morrey BF. Treatment of infection after total knee arthroplasty by débridement with retention of the components. *J Bone Joint Surg Am*. 1990;72:1383–1390.
- Tattevin P, Crémieux AC, Pottier P, Hutten D, Carbon C. Prosthetic joint infection: when can prosthesis salvage be considered? *Clin Infect Dis*. 1999;29:292–295. doi:10.1086/520202.
- Lora-Tamayo J, Murillo O, Iribarren JA, Soriano A, Sánchez-Somolinos M, Baraia-Etxaburu JM, et al. A large multicenter study of methicillin-susceptible and methicillin-resistant *Staphylococcus aureus* prosthetic joint infections managed with implant retention. *Clin Infect Dis*. 2013;56:182–194. doi:10.1093/cid/cis746.
- Lora-Tamayo J, Senneville É, Ribera A, Bernard L, Dupon M, Zeller V, et al. The not-so-good prognosis of streptococcal periprosthetic joint infection managed by implant retention: the results of a large multicenter study. *Clin Infect Dis*. 2017;64:1742–1752. doi:10.1093/cid/cix227.
- Martínez-Pastor JC, Muñoz-Mahamud E, Vilchez F, García-Ramiro S, Bori G, Sierra J, et al. Outcome of acute prosthetic joint infections due to gram-negative bacilli treated with open debridement and retention of the prosthesis. *Antimicrob Agents Chemother*. 2009;53:4772–4777. doi:10.1128/AAC.00188-09.
- Parvizi J, Gehrke T, Chen AF. Proceedings of the International Consensus on Periprosthetic Joint Infection. *Bone Joint J*. 2013;95-B:1450–1452. doi:10.1302/0301-620X.95B1.33135.
- Byren I, Bejon P, Atkins BL, Angus B, Masters S, McLardy-Smith P, et al. One hundred and twelve infected arthroplasties treated with “DAIR” (debridement, antibiotics and implant retention): antibiotic duration and outcome. *J Antimicrob Chemother*. 2009;63:1264–1271. doi:10.1093/jac/dkp107.
- Abolins CA, Page MA, Buisson KL, Jenney AWJ, Daffy JR, Choong PFM, et al. Treatment of staphylococcal prosthetic joint infections with debridement, prosthesis retention and oral rifampicin and fusidic acid. *Clin Microbiol Infect*. 2007;13:586–591. doi:10.1111/j.1469-0691.2007.01691.x.
- Senneville E, Joulie D, Legout L, Valette M, Dezéque H, Bertrand E, et al. Outcome and predictors of treatment failure in total hip/knee prosthetic joint infections due to *Staphylococcus aureus*. *Clin Infect Dis*. 2011;53:334–340. doi:10.1093/cid/cir402.
- Duggan JM, Georgiadis G, VanGorp C, Kleshinski J. Group B streptococcal prosthetic joint infections. *J South Orthop Assoc*. 2001;10:209–214; discussion 214.
- Meehan AM, Osmon DR, Duffy MCT, Hanssen AD, Keating MR. Outcome of penicillin-susceptible streptococcal prosthetic joint infection treated with debridement and retention of the prosthesis. *Clin Infect Dis*. 2003;36:845–849. doi:10.1086/368182.
- Everts RJ, Chambers ST, Murdoch DR, Rothwell AG, McKie J. Successful antimicrobial therapy and implant retention for streptococcal infection of prosthetic joints. *ANZ J Surg*. 2004;74:210–214. doi:10.1111/j.1445-2197.2004.02942.x.
- Zeller V, Lavigne M, Biau D, Leclerc P, Ziza JM, Mamoudy P, et al. Outcome of group B streptococcal prosthetic hip infections compared to that of other bacterial infections. *Joint Bone Spine*. 2009;76:491–496. doi:10.1016/j.jbspin.2008.11.010.
- Sendi P, Christensson B, Uçkay I, Trampuz A, Achermann Y, Boggian K, et al. Group B streptococcus in prosthetic hip and knee joint-associated infections. *J Hosp Infect*. 2011;79:64–69. doi:10.1016/j.jhin.2011.04.022.

- [24] Corvec S, Illiaquer M, Touchais S, Boutoille D, van der Mee-Marquet N, Quentin R, et al. Clinical features of group B *Streptococcus* prosthetic joint infections and molecular characterization of isolates. *J Clin Microbiol*. 2011;49:380–382. doi:10.1128/JCM.00581-10.
- [25] Tornero E, Senneville E, Euba G, Petersdorf S, Rodriguez-Pardo D, Lakatos B, et al. Characteristics of prosthetic joint infections due to *Enterococcus* sp. and predictors of failure: a multi-national study. *Clin Microbiol Infect*. 2014;20:1219–1224. doi:10.1111/1469-0691.12721.
- [26] Rodriguez-Pardo D, Pigrau C, Lora-Tamayo J, Soriano A, del Toro MD, Cobo J, et al. Gram-negative prosthetic joint infection: outcome of a debridement, antibiotics and implant retention approach. A large multicentre study. *Clin Microbiol Infect*. 2014;20:0911–0919. doi:10.1111/1469-0691.12649.
- [27] Fisman DN, Reilly DT, Karchmer AW, Goldie SJ. Clinical effectiveness and cost-effectiveness of 2 management strategies for infected total hip arthroplasty in the elderly. *Clin Infect Dis*. 2001;32:419–430. doi:10.1086/318502.
- [28] Dzaja I, Howard J, Somerville L, Lanting B. Functional outcomes of acutely infected knee arthroplasty: a comparison of different surgical treatment options. *Can J Surg*. 2015 Dec;58(6):402–407.
- [29] Sherrell JC, Fehring TK, Odum S, Hansen E, Zmistowski B, Dennen A, et al. The Chitranjan Ranawat Award: fate of two-stage reimplantation after failed irrigation and débridement for periprosthetic knee infection. *Clin Orthop Relat Res*. 2011;469:18–25. doi:10.1007/s11999-010-1434-1.
- [30] Nodzo SR, Boyle KK, Nocon AA, Henry MW, Mayman DJ, Westrich GH. The influence of a failed irrigation and debridement on the outcomes of a subsequent 2-stage revision knee arthroplasty. *J Arthroplasty*. 2017;32:2508–2512. doi:10.1016/j.arth.2017.03.026.
- [31] Brimmo O, Ramanathan D, Schiltz NK, Pillai ALPC, Klika AK, Barsoum WK. Irrigation and debridement before a 2-stage revision total knee arthroplasty does not increase risk of failure. *J Arthroplasty*. 2016;31:461–446. doi:10.1016/j.arth.2015.08.044.

Authors: In Jun Koh, Adrian Taylor, Tae-Kyun Kim, Prashant Meshram

QUESTION 4: Does exchange of all modular components during debridement, antibiotic and implant retention (DAIR) reduce the rate of surgical site infection (SSI)/periprosthetic joint infection (PJI) recurrence?

RECOMMENDATION: Yes. Exchange of all the modular components during DAIR reduces the risk of PJI recurrence.

LEVEL OF EVIDENCE: Moderate

DELEGATE VOTE: Agree: 94%, Disagree: 4%, Abstain: 2% (Super Majority, Strong Consensus)

RATIONALE

Prosthetic joint infections in the early stage are commonly treated with DAIR. If successful, the outcomes of PJI treated by DAIR show functional outcomes and patient reported outcomes equivalent to those of primary total joint replacements [1]. During this procedure, the removal of modular components allows for better visualization of the knee, especially in the posterior aspect, thereby facilitating proper debridement and potential bio-burden/bio-film elimination. However, it is difficult to judge the necessity of exchanging the modular components during DAIR surgery due to the lack of conclusive evidence.

Our literature review identified several studies that support the exchange of modular components to reduce the rate of PJI recurrence [1–7]. Amongst these, six are retrospective and one is a meta-analysis [7] involving 39 retrospective case-control and cohort studies. Notably, all the studies included in this meta-analysis were also retrospective, making its strength of evidence inherently limited. Furthermore, the success rates after modular exchange during DAIR shows a wide range of variation from 18–83% among different cohorts in various studies. Such wide variations in the impact of modular component exchange suggests that the outcome of DAIR may be associated with multiple factors such as patient selection, thoroughness of debridement, type and virulence of the microorganisms, choice and duration of antibiotic regimen and the definition of treatment failure rather than the exchange of modular components itself. However, a recent systematic review [7] of DAIR performed for total hip arthroplasty showed that the mean proportion of success rate in studies where modular components were exchanged was significantly higher (73.9%) than studies in which no components were exchanged (60.7%). A multicenter review article [5] of 349 patients with *Staphylococcus aureus* PJI of both hip and knee replacements reported that modular exchange reduced the risk of failure by 33%. In addition, PJI review articles [8,9] and Choi et al. [2] study suggest that in total knee arthroplasty, not exchanging the polyethylene was an independent predictor of failure of DAIR (100% failure

versus 59% success with modular exchange). Moreover, a recent case-controlled study [3] has shown the ten year implant survival rate of 86% with modular component exchange in DAIR (as compared to 68% without modular exchange) along with a fourfold increase in eradication rate. In contrast, there are several other studies which suggest that modular component exchange is not related to higher success rate of DAIR [8,10–15].

Due to the lack of conclusive evidence in the form of well-designed prospective randomized trials and standardized protocols, only a moderate strength of recommendation is provided for exchanging the modular components during DAIR to reduce the PJI recurrence rate.

REFERENCES

- [1] Grammatopoulos G, Bolduc ME, Atkins BL, Kendrick BJL, McLardy-Smith P, Murray DW, et al. Functional outcome of debridement, antibiotics and implant retention in periprosthetic joint infection involving the hip: a case-control study. *Bone Joint J*. 2017;99-B:614–622. doi:10.1302/0301-620X.99B5.BJJ-2016-0562.R2.
- [2] Choi H-R, von Knoch F, Zurakowski D, Nelson SB, Malchau H. Can implant retention be recommended for treatment of infected TKA? *Clin Orthop Relat Res*. 2011;469:961–969. doi:10.1007/s11999-010-1679-8.
- [3] Grammatopoulos G, Kendrick B, McNally M, Athanasou NA, Atkins B, McLardy-Smith P, et al. Outcome following debridement, antibiotics, and implant retention in hip periprosthetic joint infection- an 18-year experience. *J Arthroplasty*. 2017;32:2248–2255. doi:10.1016/j.arth.2017.02.066.
- [4] Kim JG, Bae JH, Lee SY, Cho WT, Lim HC. The parameters affecting the success of irrigation and debridement with component retention in the treatment of acutely infected total knee arthroplasty. *Clin Orthop Relat Res*. 2015;7:69–76. doi:10.4055/cios.2015.7.1.69.
- [5] Lora-Tamayo J, Murillo O, Iribarren JA, Soriano A, Sánchez-Somolinos M, Baraia-Etxaburu JM, et al. A large multicenter study of methicillin-susceptible and methicillin-resistant *Staphylococcus aureus* prosthetic joint infections managed with implant retention. *Clin Infect Dis*. 2013;56:182–194. doi:10.1093/cid/cis746.
- [6] Lora-Tamayo J, Senneville É, Ribera A, Bernard L, Dupon M, Zeller V, et al. The not-so-good prognosis of streptococcal periprosthetic joint infection managed by implant retention: the results of a large multicenter study. *Clin Infect Dis*. 2017;64:1742–1752.
- [7] Tsang STJ, Ting J, Simpson AHRW, Gaston P. Outcomes following debridement, antibiotics and implant retention in the management of peripros-

- thetic infections of the hip: a review of cohort studies. *Bone Joint J.* 2017;99-B:1458–1466. doi:10.1302/0301-620X.99B11.BJ-2017-0088.R1.
- [8] Qasim SN, Swann A, Ashford R. The DAIR (debridement, antibiotics and implant retention) procedure for infected total knee replacement – a literature review. *SICOT J.* 2017;3:2. doi:10.1051/sicotj/2016038.
- [9] Kuiper JW, Willink RT, Moojen DJF, van den Bekerom MP, Colen S. Treatment of acute periprosthetic infections with prosthesis retention: review of current concepts. *World J Orthop.* 2014;5:667–676. doi:10.5312/wjo.v5.i5.667.
- [10] Achermann Y, Stasch P, Preiss S, Lucke K, Vogt M. Characteristics and treatment outcomes of 69 cases with early prosthetic joint infections of the hip and knee. *Infection.* 2014;42:511–519. doi:10.1007/s15010-014-0584-6.
- [11] Bryan AJ, Abdel MP, Sanders TL, Fitzgerald SF, Hanssen AD, Berry DJ. Irrigation and debridement with component retention for acute infection after hip arthroplasty: improved results with contemporary management. *J Bone Joint Surg Am.* 2017;99:2011–2018.
- [12] Deirmengian C, Greenbaum J, Lotke PA, Booth RE, Lonner JH. Limited success with open debridement and retention of components in the treatment of acute *Staphylococcus aureus* infections after total knee arthroplasty. *J Arthroplasty.* 2003;18:22–26.
- [13] Koh JJ, Han SB, In Y, Oh KJ, Lee DH, Kim TK, et al. Open debridement and prosthesis retention is a viable treatment option for acute periprosthetic joint infection after total knee arthroplasty. *Arch Orthop Trauma Surg.* 2015;135:847–855. doi:10.1007/s00402-015-2237-3.
- [14] Peel TN, Buisson KL, Dowsey MM, Aboltins CA, Daffy JR, Stanley PA, et al. Outcome of debridement and retention in prosthetic joint infections by methicillin-resistant staphylococci, with special reference to rifampin and fusidic acid combination therapy. *Antimicrob Agents Chemother.* 2013;57:350–355. doi:10.1128/AAC.02061-12.
- [15] Tornero E, Morata L, Martinez-Pastor JC, Bori G, Climent C, García-Velez DM, et al. KLIC-score for predicting early failure in prosthetic joint infections treated with debridement, implant retention and antibiotics. *Clin Microbiol Infect.* 2015;21:786.e9–786.e17. doi:10.1016/j.cmi.2015.04.012.

Authors: Wayne G. Paprosky, Evan Schwechter, Linda I. Suleiman, Jeremy Loloi, Foster Chen

QUESTION 5: What is the minimum necessary volume of irrigation solution to use in debridement, antibiotics and implant retention (DAIR) treatment of acute periprosthetic joint infection (PJI)?

RECOMMENDATION: We recommend that 6-9L of irrigation solution, including saline or antiseptic solution such as sterile dilute povidone-iodine, is used during DAIR treatment of acute PJI.

LEVEL OF EVIDENCE: Consensus

DELEGATE VOTE: Agree: 90%, Disagree: 7%, Abstain: 3% (Super Majority, Strong Consensus)

RATIONALE

To date, there are no reported clinical studies relating to the optimal volume of irrigation required during DAIR treatment of PJI. However, variable outcomes have been reported with different institutions employing individual protocols for volumes of irrigation.

Few studies provide limited secondary data with regards to the ideal volume of irrigation to be used during total joint arthroplasty (TJA) in general and treatment of an infected joint in particular. In one such study, the authors were able to determine that four liters of sterile saline pulse lavage were sufficient to remove bone and polymethyl methacrylate (PMMA) debris exceeding 1µm in size from the joint during TJA. The authors extrapolated from their results that bacteria might effectively be removed with the same amount of irrigation given the similarity in size to the particulates assessed [1]. This model did not consider the effect of the developing bacterial biofilm on infected arthroplasty implants. DAIR has traditionally been thought to reduce the bacterial load and be effective in the acute period given that bacteria theoretically had not yet formed a glyco-calyx biofilm. In another study, the authors used an in vitro model to determine the efficacy of biofilm removal from arthroplasty implants using high-pressure pulsatile lavage. Three liters of normal saline were used over an area measuring 1cm² recreating a prosthesis covered in *Staphylococcus aureus* biofilm. The authors concluded that pulse lavage is not able to sufficiently debride pre-existing biofilm. The volume of irrigation solution required was not investigated as a primary endpoint and the authors caution against extrapolating the results to clinical scenarios as their in vitro model potentially overestimated the amount of biofilm debrided by three liters of sterile saline pulse lavage [2]. More important than the volume of irrigation, researchers have found that the presence of staphylococcal infection, an elevated American Society of Anesthesiologists (ASA) score, or purulence were more likely to determine failure.

A comprehensive systematic review of the literature relating to open DAIR treatment of acute postoperative and hematogenous periprosthetic hip and/or knee joint infections, with or without modular component exchange, was performed. Databases searched include: PubMed, Embase, Cochrane Review and Google Scholar. Initial query generated 664 articles. Review articles and book chapters were excluded, while all references from such sources were screened for inclusion (spanning from 1990-2017). We included all Level I-IV studies that specified a certain volume of irrigation used per procedure and recorded the type of solution(s) used, mode of lavage administration, use of additive(s) and number of irrigation and debridements (I&Ds) performed. We included cases whereby some of the modular components may have been exchanged, but excluded those with dedicated planned staged exchanges. A total of 14 studies met the aforementioned criteria (Table 1) [3–16].

Typically, around 6 to 9L of solution were used during a single DAIR treatment, with 12 of the 14 studies utilizing up to 9L or more of irrigation solution. The evidence base for the specific irrigation volume is poorly defined within all studies, and recommendations for specific volumes in both primary and review articles reference consensus data obtained from previously published guidelines or individual protocols. [17–22] Therefore, this systematic review represents the body of evidence of actual irrigation volumes reportedly used in the literature.

No studies currently exist directly linking the necessary volume of irrigation to use in DAIR in acute PJI. Based on several retrospective studies, we extrapolate that the use of 6-9L of irrigation solution may be required when treating acute PJI. Prospective studies evaluating the volume of irrigation used as a study endpoint are required to better elucidate the optimal volume of irrigation in DAIR treatment of PJI.

TABLE 1. DAIR studies

Reference (Author, Year)	Study Design	n (acutePJIs)	Irrigation Solution	Additives	Volume Per Procedure (L)	Modular Revision	Infections Controlled
Mont et al (1997)	Prospective	24	NS	None	10	Yes	83%
Azzam et al (2010)	Retrospective	104	NS	Antibiotics	9	Some	44%
Estes et al (2010)	Retrospective	20	Castile soap solution	None	6 to 9	Yes	90%
Koyonos et al (2011)	Retrospective	102	NS	Antibiotics	9	No	35%
Royo et al (2013)	Retrospective	34	NS	Betadine/Peroxide	9	Some	74%
Kim et al (2014)	Retrospective	20	NS	Betadine	6 to 9	Yes	100%
Moojen et al (2014)	Retrospective	68	NS	None	3 to 6	Yes	21%
Koh et al (2015)	Retrospective	52	NS	None	9	Some	71%
Sousa et al (2016)	Prospective	23	NS	Chlorhexidine	7	Yes	85%
Tornero et al (2016)	Retrospective	143	Sterile Water	None	6 to 9	No	88%
Bryan et al (2017)	Retrospective	90	NS	None	6 to 9	Some	87%
Di Benedetto et al (2017)	Retrospective	20	NS	Betadine	6 to 9	Yes	80%
Duque et al (2017)	Retrospective	67	NS	Betadine/Dakin's/Bacitracin	12	Yes	69%
Narayanan et al (2017)	Retrospective	55	N/A	None	9	Yes	60%

REFERENCES

- [1] Niki Y, Matsumoto H, Otani T, Tomatsu T, Toyama Y. How much sterile saline should be used for efficient lavage during total knee arthroplasty? Effects of pulse lavage irrigation on removal of bone and cement debris. *J Arthroplasty*. 2007;22:95–99. doi:https://doi.org/10.1016/j.arth.2006.02.078.
- [2] Urish KL, DeMuth PW, Craft DW, Haider H, Davis CM 3rd. Pulse lavage is inadequate at removal of biofilm from the surface of total knee arthroplasty materials. *J Arthroplasty*. 2014;29:1128–1132. doi:10.1016/j.arth.2013.12.012.
- [3] Azzam KA, Seeley M, Ghanem E, Austin MS, Purtill JJ, Parvizi J. Irrigation and debridement in the management of prosthetic joint infection: traditional indications revisited. *J Arthroplasty*. 2010;25:1022–1027. doi:10.1016/j.arth.2010.01.104.
- [4] Bryan AJ, Abdel MP, Sanders TL, Fitzgerald SF, Hanssen AD, Berry DJ. Irrigation and debridement with component retention for acute infection after hip arthroplasty: improved results with contemporary management. *J Bone Joint Surg Am*. 2017;99:2011–2018.
- [5] Di Benedetto P, Di Benedetto ED, Salviato D, Beltrame A, Gisonni R, Cainero V, et al. Acute periprosthetic knee infection: Is there still a role for DAIR? *Acta Biomedica*. 2017;88:84–91. doi:10.23750/abm.v88i2-S.6518.
- [6] Duque AF, Post ZD, Lutz RW, Orozco FR, Pulido SH, Ong AC. Is there still a role for irrigation and debridement with liner exchange in acute periprosthetic total knee infection? *J Arthroplasty*. 2017;32:1280–1284.
- [7] Estes CS, Beauchamp CP, Clarke HD, Spangehl MJ. A two-stage retention debridement protocol for acute periprosthetic joint infections. *Clin Orthop Relat Res*. 2010;468:2029–2038. doi:10.1007/s11999-010-1293-9.
- [8] Kim JH, Chun SK, Yoon YC, Lakhotia D, Shon WY. Efficacy of debridement for early periprosthetic joint infection after hip arthroplasty. *Hip Pelvis*. 2014;26:227–234. doi:10.5371/hp.2014.26.4.227.
- [9] Koh JJ, Han SB, In Y, Oh KJ, Lee DH, Kim TK. Open debridement and prosthesis retention is a viable treatment option for acute periprosthetic joint infection after total knee arthroplasty. *Arch Orthop Trauma Surg*. 2015;135:847–855. doi:10.1007/s00402-015-2237-3.
- [10] Koyonos L, Zmistowski B, Della Valle CJ, Parvizi J. Infection control rate of irrigation and debridement for periprosthetic joint infection. *Clin Orthop Relat Res*. 2011;469:3043–3048. doi:10.1007/s11999-011-1910-2.
- [11] Moojen DJF, Zwiers JH, Scholtes VA, Verheyen CC, Poolman RW. Similar success rates for single and multiple debridement surgery for acute hip arthroplasty infection. *Acta Orthopaedica*. 2014;85:383–388. doi:10.3109/17453674.2014.927729.
- [12] Mont MA, Waldman B, Banerjee C, Pacheco IH, Hungerford DS. Multiple irrigation, debridement, and retention of components in infected total knee arthroplasty. *J Arthroplasty*. 1997;12:426–433. doi:https://doi.org/10.1016/S0883-5403(97)90199-6.
- [13] Narayanan R, Anoushiravani AA, Elbuluk AM, Chen KK, Adler EM, Schwarzkopf R. Irrigation and debridement for early periprosthetic knee infection: is it effective? *J Arthroplasty*. 2018. doi:10.1016/j.arth.2017.12.039.
- [14] Royo A, Bertrand ML, Ramos L, Fernandez-Gordillo F, Guaredo E. Is there still a place for continuous closed irrigation in the management of periprosthetic total knee infection? *Open Orthop J*. 2013;7:205–210.
- [15] Tornero E, Morata L, Martinez-Pastor JC, Angulo S, Combalia A, Bori G, et al. Importance of selection and duration of antibiotic regimen in prosthetic joint infections treated with debridement and implant retention. *J Antimicrob Chemother*. 2016;71:1395–1401. doi:10.1093/jac/ckv481.
- [16] Laffer RR, Ruef C. Diagnosis and treatment of prosthetic joint infections. *Z Rheumatol*. 2006;65(1):12,14–17.
- [17] Jiranek WA, Waligora AC, Hess SR, Golladay GL. Surgical treatment of prosthetic joint infections of the hip and knee: changing paradigms? *J Arthroplasty*. 2015;30:912–918. doi:10.1016/j.arth.2015.03.014.
- [18] Fink B, Schuster P, Schwenninger C, Frommelt L, Oremek D. A standardized regimen for the treatment of acute postoperative infections and acute hematogenous infections associated with hip and knee arthroplasties. *J Arthroplasty*. 2017;32:1255–1261. doi:10.1016/j.arth.2016.10.011.
- [19] Martínez-Pastor JC, Maculé-Beneyto F, Suso-Vergara S. Acute infection in total knee arthroplasty: diagnosis and treatment. *Open Orthop*. 2013;7:197–204. doi:10.2174/1874325001307010197.
- [20] Volpin A, Sukeik M, Alazzawi S, Haddad FS. Aggressive early debridement in treatment of acute periprosthetic joint infections after hip and knee replacements. *Open Orthop*. 2016;10:669–678. doi:10.2174/1874325001610010669.
- [21] Parvizi J, Cavanaugh PK, Diaz-Ledezma C. Periprosthetic knee infection: ten strategies that work. *Knee Surg Relat Res*. 2013;25:155–164. doi:10.5792/ksrr.2013.25.4.155.
- [22] Gehrke T, Alijanipour P, Parvizi J. The management of an infected total knee arthroplasty. *Bone Joint J*. 2015;97-B:20–29. doi:10.1302/0301-620X.97B10.36475.



Authors: Leo Whiteside, Briande Beaubien, Kimberly E. Martin, Christopher Ferry

QUESTION 6: Is there a role for direct intra-articular antibiotic infusion following irrigation and debridement (I&D) for periprosthetic joint infection (PJI)?

RECOMMENDATION: The concept of achieving a minimum biofilm eradication concentration (MBEC) of antibiotics at the site of the infection is compelling. Despite the presence of retrospective studies reporting favorable outcome, because of heterogeneity in terms of adjunctive antibiotics, absence of a control group and small cohort size, the routine administration of intra-articular antibiotics in treatment of PJI is not justified. Prospective, randomized controlled trials (RCTs) are needed to support the routine use of intra-articular antibiotics as a stand-alone or adjunct treatment of PJI.

LEVEL OF EVIDENCE: Consensus

DELEGATE VOTE: Agree: 92%, Disagree: 6%, Abstain: 2% (Super Majority, Strong Consensus)

RATIONALE

Current published evidence for intra-articular antibiotic infusion following irrigation and debridement for PJI is limited to small case series and retrospective cohort studies. The authors of all studies aimed to achieve higher concentrations of antibiotics at the site of the infection than is possible with systemic therapy. PJI is associated with the presence of biofilms and sessile bacteria that are encapsulated within a biofilm matrix are more difficult to eradicate than planktonic bacteria [1–7]. Biofilm is the single most important factor causing resistance of bacteria to antibiotics in the treatment of PJI. While modest antibiotic concentration can prevent biofilm formation, eliminating established biofilm is a different matter. Bacteria protected by biofilm requires concentrations that are orders of

magnitude greater than the minimal inhibitory concentration for the planktonic forms of the same bacterium to eliminate resistant organisms that are protected by the glycocalyx.

A systematic review of the literature revealed that biofilm encapsulated bacteria requires MBEC of antibiotics that are several orders of magnitude (100–1000+) above the minimum inhibitory concentrations (MIC) sufficient to eradicate planktonic bacteria (Table 1). Currently, MBECs at the site of the joint infection are not achievable with traditional intravenous (IV) antibiotic therapy without systemic toxicity (Table 1). IV antibiotics generally do not achieve these levels of concentration in synovial fluid, but instead achieve levels around two to three times the MIC.

Even though extensive work has been done to develop adjuvant agents such as antibacterial peptides and chelating agents to reduce the resistance of biofilm bacteria to antibiotics, the only clinically viable method available now is to apply antibiotics directly to the affected joint where the implant resides to achieve concentrations high enough to approach MBEC. The use of antibiotic-impregnated polymethyl methacrylate spacers is the most common method used to deliver antibiotics directly into the joint as part of treatment of PJI. While intra-articular concentration of antibiotics is significantly higher when antibiotic-loaded spacers are used, the level is still an order of magnitude (perhaps thousands of times) lower than what is needed to eradicate the biofilm. Local delivery of antibiotics with antibiotic-laden bone cement does not apply a consistent dose for enough time, with most the elution occurring in the first 48-72 hours and by day 5, the concentrations are often sub-therapeutic [8]. Time is an important factor in management of biofilm and exposure to high concentrations for long periods enhances the ability to achieve MBEC.

Direct antibiotic infusion through an infusion pump can achieve extremely high local levels of antibiotics for a prolonged period. In addition, when the antibiotic is delivered through an external portal, it can be discontinued if toxicity or sensitivity occurs. Perry et al. were the first group to describe intra-articular instillation of antibiotics in 1992 [9]. They used an implantable pump with a catheter from the wound surface, to deliver 200-350 mg of amikacin in a 50mg/ml dilution for 8-15 weeks, to 72 patients with acute infections. Of these patients, 49 underwent debridement and retained their prostheses and 23 had their prostheses removed after the initial debridement. They only reported in detail on a subset of 12 patients (10 knees and 2 hips, median age of 59) with no prior history of infection and with a 37-month follow-up. Local levels of antibiotics were assessed by assaying wound drainage or synovial fluid and ranged from 150 ug/ml to 1688 ug/ml. Serum levels were 10ug/ml, except for one patient whose serum concentration rose to 13ug/ml. Two patients developed recurrent infection, one with the same organism *Staphylococcus aureus* (*S. aureus*) and the other patient was infected with *Staphylococcus epidermidis*, after originally infected with *S. aureus*. In the series of 49 patients who retained their prostheses, 38 patients were infection free, however, follow-up times ranged from 1-58 months.

Fukagawa et al. reported on their experience with 15 patients (16 knees) treated for PJI with stable prostheses [10]. A causative micro-organism was identified in eight patients. Patients were treated with open synovectomy, debridement, exchange of polyethylene insert and retained their implant. In the five patients with tumor megaprotheses, the anchors were retained. A Hickman catheter was inserted percutaneously and organism specific antibiotics (if an organism was cultured) were infused into the joint space twice per day until clinical signs of infection resolved, and white blood cell (WBC) count, C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR) normalized, at which point the catheters were pulled. The mean infusion duration was 20.8 days $\pm$ 11.7 days. Intra-articular antibiotics used were: amikacin (400 mg/day), gentamicin (80mg/day) and arbekacin (200 mg/day). No serum antibiotic levels were reported. All patients also received IV or oral antibiotic therapy for 1-3 months. All patients were considered infection free and clinically healed during the first follow-up period of 46.7 months ($\pm$ 25.7 months). However, four of the five knees treated with tumor megaprotheses developed recurrent infection after a mean of 28.3 ($\pm$ 26.1 months). These patients were treated with intra-articular antibiotics again for 13-22 days and the infection was clear at last follow-up. No local toxicity or infection at the catheter site was reported.

Tsumura et al. [11] reported on the treatment of early knee PJI in ten patients with continuous, concentrated, antibiotic irrigation for 7-29 days. Antibiotics were administered through a Salem double lumen catheter after debridement with implant retention. Eight of the 10 patients were infection free and able to retain the original prostheses. The two failures were the only patients with methicillin-resistant *Staphylococcus aureus* (MRSA). Antibiotics administered were: clindamycin, amikacin, cefotiam, imipenem, arbekacin, piperacillin, cefazolin, ampicillin and vancomycin. No serum or synovial antibiotic levels were reported.

In two recent publications, Whiteside et al. reported on a retrospective cohort of 18 total knee arthroplasty (TKA) patients with recurrent knee PJIs treated with single-stage (10 patients) or two-stage revision arthroplasty (8 patients), including 3 patients that required limb lengthening and soft tissue expansion [12,13]. Intra-articular antibiotic infusion using a Hickman catheter was performed as an adjunct to meticulous debridement. The authors administered 100 mg of vancomycin or 20 mg of gentamicin in 3 mL of saline into the joint space and increased the dosage to 500 mg of vancomycin or 80 mg of gentamicin in 8 mL of saline, every 12 or 24 hours as tolerated, once the wound was stable and dry. Patients were also treated postoperatively with 1 gm of IV vancomycin and 80 mg of IV gentamicin for 48 hours. The intra-articular antibiotics were continued for six weeks, with intra-articular vancomycin levels ranging from 10,233- 20,167 mg/L. Mean serum vancomycin peak and trough levels were 4.1 $\pm$ 1.2 μ g/mL and 3.3 $\pm$ 1.2 μ g/mL respectively. Three patients had to have a reduction in the antibiotic dose due to excessive rise in the level of antibiotics. Follow-up ranged from 2.3-12 years, with a mean of 6.1 years. One patient had a recurrent, postoperative infection at 13 months. No other patients had clinical or serological signs of infection and no patient was placed on chronic suppressive antibiotics. Similarly, Roy et al. compared synovial concentrations of antibiotics with IV vs. intra-articular administration in a subset of patients in the Whiteside study cohort, and found an average, peak intra-articular vancomycin concentration of 9,242 $\pm$ 7,608 mg/L following intra-articular antibiotic infusion compared to an average intra-articular concentration of 6.8 μ g/mL following IV administration [14]. These data suggest with reasonable certainty that direct intra-articular infusion of antibiotics offers a significant benefit in treating resistant organisms, but certainly do not rise to the same level of evidence as would a RCT performed at the same center.

Revision after reinfected, two-stage revision total joint arthroplasty is an especially challenging clinical problem and is even more difficult when multiple failures have occurred. The complication rate of using antibiotic spacers is substantial including dislocation, fracture and migration of the spacer with bone loss that must be considered when contemplating a second two-stage exchange procedure. A revision with intra-articular antibiotic infusion may play a role in this scenario to reduce morbidity. Antony et al. described intra-articular antibiotic infusion as an adjunct to single-stage revision for previously failed single- or two-stage revision for knee, hip or shoulder PJI, in 57 patients with a mean age of 65 years [15]. Hickman catheters were used for intra-articular infusion of organism specific antibiotics for approximately 4-6 weeks, once or twice per day without concomitant systemic antibiotics. The intra-articular antibiotic dose administered was determined to be 50% of the serum dose given the enclosed space. Infection eradication was defined as negative culture, and normal ESR and CRP and 89.5% of patients were successfully treated at 11 months follow-up. Synovial levels of antibiotics were not collected.

TABLE 1. Therapeutic range, toxicity, minimum biofilm eradication concentration (MBEC), and minimum inhibitory concentration (MIC) of antibiotics used to treat biofilm-encapsulated bacteria

Antibiotic	Therapeutic Range	Toxic Plasma Concentration	<i>S. aureus</i>		<i>MRSA</i>		<i>P. aeruginosa</i>		<i>S. epidermidis</i>		<i>E. coli</i>	
			MIC	MBEC	MIC	MBEC	MIC	MBEC	MIC	MBEC	MIC	MBEC
Azithromycin	0.04-1	-			512	5120		2560				
Ceftazidime	<150	-					1-4	2560-5120				
Ciprofloxacin	2.5-4	11.5			0.06- >32	256-1280	0.25-2	80-1280				
Clindamycin	<0.5	-			0.015-0.06	64->1024						
Colistin	1-4	-										
Daptomycin	6-10	-	0.25	600	0.125	1014						
Doxycycline	<10	30			0.064-0.125	64-128		2560				
Erythromycin	0.5-6	12-15	1	6400	0.12->256	64->1024		512xMIC				
Gentamicin	5-10	12	1	6400	0.06-64	1->256						
Linezolid	0.5-4	-	1	6400	1-2	4->1024	4-128	>5120				
Piperacillin	5-20	-										
Rifampicin	0.1-10	204	0.16	40								
Tobramycin	5-10	12-15	1	160-4000	1	≥8000	0.2-16	250-2560	32	≥8000	2	62.5-125
Vancomycin	<5-10	30	2	2000-8000	0.25-2	2000-8000			2	1000-8000		

REFERENCES

- [1] Abdi-Ali A, Mohammadi-Mehr M, Agha Alaei Y. Bactericidal activity of various antibiotics against biofilm-producing *Pseudomonas aeruginosa*. *Int J Antimicrob Agents*. 2006;27:196–200. doi:10.1016/j.ijantimicag.2005.10.007.
- [2] Castaneda P, McLaren A, Tavaziva G, Overstreet D. Biofilm antimicrobial susceptibility increases with antimicrobial exposure time. *Clin Orthop Relat Res*. 2016;474:1659–1664. doi:10.1007/s11999-016-4700-z.
- [3] Dosler S, Karaaslan E. Inhibition and destruction of *Pseudomonas aeruginosa* biofilms by antibiotics and antimicrobial peptides. *Peptides*. 2014;62:32–37. doi:10.1016/j.peptides.2014.09.021.
- [4] Goel S, Mishra P. Thymoquinone inhibits biofilm formation and has selective antibacterial activity due to ROS generation. *Appl Microbiol Biotechnol*. 2018;102:1955–1967. doi:10.1007/s00253-018-8736-8.
- [5] Regenthal R, Krueger M, Koepfel C, Preiss R. Drug levels: therapeutic and toxic serum/plasma concentrations of common drugs. *J Clin Monit Comput*. 1999;15:529–544.
- [6] Schulz M, Iwersen-Bergmann S, Andresen H, Schmoldt A. Therapeutic and toxic blood concentrations of nearly 1,000 drugs and other xenobiotics. *Crit Care*. 2012;16:R136. doi:10.1186/cc11441.
- [7] Marques C, Tasse J, Pracros A, Collin V, Franceschi C, Laurent F, et al. Effects of antibiotics on biofilm and unattached cells of a clinical *Staphylococcus aureus* isolate from bone and joint infection. *J Med Microbiol*. 2015;64:1021–1026. doi:10.1099/jmm.0.000125.
- [8] Kuehle DK, Landon GC, Musher DM, Noble PC. Elution of vancomycin, daptomycin, and amikacin from acrylic bone cement. *Clin Orthop Relat Res*. 1991;302–308.
- [9] Perry CR, Hulsey RE, Mann FA, Miller GA, Pearson RL. Treatment of acutely infected arthroplasties with incision, drainage, and local antibiotics delivered via an implantable pump. *Clin Orthop Relat Res*. 1992;216–223.
- [10] Fukagawa S, Matsuda S, Miura H, Okazaki K, Tashiro Y, Iwamoto Y. High-dose antibiotic infusion for infected knee prosthesis without implant removal. *J Orthop Sci*. 2010;15:470–476. doi:10.1007/s00776-010-1487-8.
- [11] Tsumura H, Ikeda S, Ono T, Itonaga I, Taira H, Torisu T. Synovectomy, debridement, and continuous irrigation for infected total knee arthroplasty. *Int Orthop*. 2005;29:113–116. doi:10.1007/s00264-004-0626-2.
- [12] Whiteside LA, Peppers M, Nayfeh TA, Roy ME. Methicillin-resistant *Staphylococcus aureus* in TKA treated with revision and direct intra-articular antibiotic infusion. *Clin Orthop Relat Res*. 2011;469:26–33. doi:10.1007/s11999-010-1313-9.
- [13] Whiteside LA, Nayfeh TA, LaZear R, Roy ME. Reinfected revised TKA resolves with an aggressive protocol and antibiotic infusion. *Clin Orthop Relat Res*. 2012;470:236–243. doi:10.1007/s11999-011-2087-4.
- [14] Roy ME, Peppers MP, Whiteside LA, LaZear RM. Vancomycin concentration in synovial fluid: direct injection into the knee vs. intravenous infusion. *J Arthroplasty*. 2014;29:564–568. doi:10.1016/j.arth.2013.08.017.
- [15] Antony SJ, Westbrook RS, Jackson JS, Heydemann JS, Nelson JL. Efficacy of single-stage revision with aggressive debridement using intra-articular antibiotics in the treatment of infected joint prosthesis. *Infect Dis (Auckl)*. 2015;8:17–23. doi:10.4137/IDRT.S26824.

● ● ● ● ●

Authors: Rafael J. Sierra, George Babis, Jean Noël Argenson

QUESTION 7: Can debridement, antibiotics and implant retention (DAIR) be utilized in patients with an acute chronic infection of a unicompartamental knee arthroplasty (UKA)?

RECOMMENDATION: In the event of acute infection following UKA, early DAIR can be considered. However, if initial treatment effort results in failure or chronic infection is present, the implanted prosthesis should be removed and a one-stage or two-stage conversion to total knee arthroplasty (TKA) should be performed in combination with antibiotic therapy.

LEVEL OF EVIDENCE: Limited

DELEGATE VOTE: Agree: 96%, Disagree: 2%, Abstain: 2% (Unanimous, Strongest Consensus)

RATIONALE

The main reasons for revision of UKA are loosening, progression of osteoarthritis to another compartment and infection [1]. The incidence of infection after UKA at 0.2 to 1% is lower than that reported after total knee arthroplasty (TKA) [1,2]. A distinctive feature of UKA infection is that both the prostheses and the native cartilage are involved [1]. This is in part attributed to the use of minimally invasive exposures, with less damage to the adjacent soft tissue and sparing of bone and ligamentous structures [3].

In the event of immediate or acute infection following UKA, early irrigation and debridement followed by antibiotic adminis-

tration can be a proper treatment solution. However, if the initial treatment effort ends up in failure or chronic infection is present, the implanted prosthesis should be removed and a one- or two-stage revision surgery should be carried out [3]. Labruey et al. reported on failures for nine infected UKA cases managed with one-stage irrigation, debridement and conversion to TKA in combination with three months of antibiotic therapy [1]. Of note, five of these cases first failed DAIR. Kim et al. reported management of five infected UKA cases with two-stage conversion to TKA [3]. Bohm et al. reported two infected UKAs, one of which was managed with one-stage conversion

TABLE 1. Summary of infected UKA cases in the literature

Author/Year	N (infected UKA cases)	Failed DAIR	Treatment	Failures	Follow-up
Labruey 2015[1]	9	5	one-stage conversion to TKA (9)	0	Median 60 months
Bohm 2000[4]	2 (0.7% infection rate)	?	one-stage (1) two-stage (1)	1 (AKA)	Mean 4 years
Saragaglia 2013[5]	8 (2% of failed UKAs)	?	?	?	?
Kim 2016[3]	5 (0.3% infection rate)	?	two-stage (5)	?	?

successfully and the other was treated with two-stage conversion, ultimately resulting in above the knee amputation [4].

In the setting of UKA, recommendations are weak as only five published papers examine the results of failed UKA, including infection and the rate of infection is very low (Table 1). Two of the infected UKA cases in one study [1] had been post-traumatic infections prior to implantation of the UKA and thus represent more complex scenarios potentially predisposing to treatment failure. There is no literature directly evaluating the role of DAIR in the setting of UKA. However, subsequent failure due to progression of osteoarthritis (OA) occurred in two cases (survival 49%) at an average of three years. Therefore, it may be advisable to proceed with one- or two-stage conversion to TKA at the time of infection in the setting of UKA to minimize the need for additional revision procedures in the future and prevent associated morbidity.

In general, the surgeon should assess prior UKA function, component position and fixation and condition of alternate knee

compartments to determine whether retention of implants with DAIR is an appropriate initial treatment in the setting of infection.

REFERENCES

- [1] Labruière C, Zeller V, Lhotellier L, Desplaces N, Léonard P, Mamoudy P, et al. Chronic infection of unicompartmental knee arthroplasty: one-stage conversion to total knee arthroplasty. *Orthop Traumatol Surg Res.* 2015;101:553–557. doi:10.1016/j.otsr.2015.04.006.
- [2] Vasso M, Corona K, D'Apolito R, Mazzitelli G, Panni AS. Unicompartmental knee arthroplasty: modes of failure and conversion to total knee arthroplasty. *Joints.* 2017;5:44–50. doi:10.1055/s-0037-1601414.
- [3] Kim KT, Lee S, Lee JI, Kim JW. Analysis and treatment of complications after unicompartmental knee arthroplasty. *Knee Surg Relat Res.* 2016;28:46–54. doi:10.5792/ksrr.2016.28.1.46.
- [4] Böhm I, Landsiedl F. Revision surgery after failed unicompartmental knee arthroplasty: a study of 35 cases. *J Arthroplasty.* 2000;15:982–989.
- [5] Saragaglia D, Bonnin M, Dejour D, Deschamps G, Chol C, Chabert B, et al. Results of a French multicentre retrospective experience with four hundred and eighteen failed unicompartmental knee arthroplasties. *Int Orthop.* 2013;37:1273–1278. doi:10.1007/s00264-013-1915-4.



Authors: Dwikora Novembri Utomo, Nicolaas Budhiparama, Andrew Battenberg, Ferdiansyah Mahyudin, KuKuh Dwiputra Hernugrahanto, I. Lumban-Gaol

QUESTION 8: Can debridement, antibiotics and implant retention (DAIR) be utilized in the treatment of acute periprosthetic joint infection (PJI) with a megaprosthesis?

RECOMMENDATION: DAIR is a viable treatment option in acute PJI of a megaprosthesis. The effectiveness of DAIR is still unclear due to lack of comparative data among the treatment options and limited evidence to suggest superiority of any one treatment. The treatment decision must be made on a case-by-case basis and account for underlying medical conditions, infection history, organism characteristics and surgical history. DAIR is most appropriate for acute PJI without complicating factors, such as extensive and pervasive infection by a high virulence or resistant organism.

LEVEL OF EVIDENCE: Limited

DELEGATE VOTE: Agree: 96%, Disagree: 1%, Abstain: 3% (Unanimous, Strongest Consensus)

RATIONALE

Acute PJI of megaprotheses is a terrible complication and a difficult situation for treatment [1]. Infection rates in patients with megaprotheses have been reported to range from 3% to greater than 30% [1–3]. In principle, the treatment of acute PJI with a megaprosthesis is similar to treatment of other acute PJI, except there is significantly more potential space and a greater soft tissue infectious burden requiring more extensive exposure and debridement [4,5]. The surgical options include DAIR [6–8], one-stage revision surgery [4], two-stage revision with an interval cement spacer [9–11], arthrodesis and amputation [5,8]. Unfortunately, there is limited data on the outcome of these different procedures [1,9]. The lack of comparative data is due to the limited indications for a megaprosthesis as well as the clinical heterogeneity of the affected patients [5]. Additionally, treatment details vary greatly, particularly for DAIR. Specific information on the debridement, the type of irrigation solutions, modular component exchange and local and systemic antibiotic use and duration are generally lacking.

Two-stage revision remains the preferred method for treatment of PJI [8–10]. However, two-stage revision significantly increases surgical and perioperative risks and includes a substantial period of reduced mobility between stages, which has heightened interest in alternative surgical options such as DAIR. DAIR is an attractive option as it may prevent the unnecessary removal of implants, which could result in further bone loss and fracture [6,11,12]. DAIR is also the simpler and less costly procedure with a demonstrated

shorter length of hospital stay [13]. The overall goal of attempting DAIR should be to select the cohort of patients in whom successful treatment is most likely.

Sujith et al. summarized the absolute and relative contraindication for DAIR [13]: The absolute contraindications are loose prosthesis, poor soft tissue coverage and compromised bone cement mantle. The relative contraindications are the presence of sinus tracts, methicillin-resistant and methicillin-susceptible *Staphylococcus aureus* (MRSA and MSSA) infection, previously revised joints, immunosuppression, rheumatoid arthritis, polymicrobial involvement, bacteremia, C-reactive protein (CRP) >100 mg/L, erythrocyte sedimentation rate (ESR) >60 mm/h, two or more previous debridements and >3 weeks of symptoms.

The decision to perform DAIR can also be based on the classification of the infection. According to Pilge et al. if intraoperative cultures are positive without other signs of infection (Tsukayama Type I), implant retention is attempted and prolonged systemic antibiotic treatment is recommended. Implant retention should also be attempted with stable arthroplasties in type II or III infections (early postoperative infection or acute hematogenous infection). If there are radiological signs of implant loosening, a one- or two-stage revision must be performed [14,15].

During DAIR, thorough debridement is necessary to improve outcome. All infected and nonviable tissue around a well-fixed prosthesis must be removed. Retained components are irrigated and

scrubbed in an effort to remove biofilm [11,13]. Various antibiotic solutions can be used intraoperatively, including dilute betadine and Dakin's solution. Culture-driven systemic antibiotics are also important for successful treatment and co-treatment with rifampin should be utilized in Staphylococcal PJIs [6]. Prolonged or chronic antibiotic suppression may also be necessary. The use of local antibiotics in addition to the administration of systemic antibiotic agents is an area of consideration. Modular components and the exposed metal of megaprotheses can be covered with antibiotic eluting cement, though there is no clinical evidence comparing the efficacy of such methods versus more simple modular exchange.

The most important factors contributing to treatment failure are longer duration of symptoms, a longer time after initial arthroplasty, the need for multiple debridements, the retention of exchangeable components and PJI caused by MRSA [6,11,12]. One- or two-stage revision should be performed if DAIR fails [11,13].

In general, DAIR is a treatment option for acute PJI with a megaprosthesis with varying levels of success in selected and non-complicated patients. The heterogeneity inherent in these cases makes comparisons difficult and there is always some degree of individualization in choice of treatment.

REFERENCES

- [1] Ercolano LB, Christensen T, McGough R, Weiss K. Treatment solutions are unclear for perimegaprosthesis infections. *Clin Orthop Relat Res*. 2013;471:3204–3213. doi:10.1007/s11999-013-2852-7.
- [2] Harges J, Gebert C, Schwappach A, Ahrens H, Streiburger A, Winkelmann W, et al. Characteristics and outcome of infections associated with tumor endoprostheses. *Arch Orthop Trauma Surg*. 2006;126:289–296. doi:10.1007/s00402-005-0009-1.
- [3] Henderson ER, Groundland JS, Pala E, Dennis JA, Wooten R, Cheong D, et al. Failure mode classification for tumor endoprostheses: retrospective review of five institutions and a literature review. *J Bone Joint Surg Am*. 2011;93:418–429. doi:10.2106/JBJS.J.00834.
- [4] Holzer G, Windhager R, Kotz R. One-stage revision surgery for infected megaprotheses. *J Bone Joint Surg Br*. 1997;79:31–35.
- [5] Kapoor SK, Thiyam R. Management of infection following reconstruction in bone tumors. *J Clin Orthop Relat Res Trauma*. 2015;6:244–251. doi:10.1016/j.jcot.2015.04.005.
- [6] Kuiper JW, Willink RT, Moojen DJF, van den Bekerom MP, Colen S. Treatment of acute periprosthetic infections with prosthesis retention: review of current concepts. *World J Orthop*. 2014;5:667–676. doi:10.5312/wjo.v5.i5.667.
- [7] Rodriguez-Pardo D, Pigrau C, Lora-Tamayo J, Soriano A, del Toro MD, Cobo J, et al. Gram-negative prosthetic joint infection: outcome of a debridement, antibiotics and implant retention approach. A large multicentre study. *Clin Microbiol Infect*. 2014;20:0911–0919. doi:10.1111/1469-0691.12649.
- [8] McDonald DJ, Fitzgerald RH, Ilstrup DM. Two-stage reconstruction of a total hip arthroplasty because of infection. *J Bone Joint Surg Am*. 1989;71:828–834.
- [9] Erallp L, Ozger H, Kocaoglu M. Treatment strategies for infected megaprosthesis. *Orthopaedic Proceedings*. 2009;91-B:301–301. doi:10.1302/0301-620X.91BSUPP_II.0910301a.
- [10] Harges J, Ahrens H, Gosheger G, Nottrott M, Dieckmann R, Henrichs MP, et al. [Management of complications in megaprotheses]. *Unfallchirurg*. 2014;117:607–613. doi:10.1007/s00113-013-2477-z.
- [11] Grammatopoulos G, Kendrick B, McNally M, Athanasou NA, Atkins B, McLardy-Smith P, et al. Outcome following debridement, antibiotics, and implant retention in hip periprosthetic joint infection - an 18-year experience. *J Arthroplasty*. 2017;32:2248–2255.
- [12] Kuiper JWP, Vos SJC, Saouti R, Vergroesen DA, Graat HCA, Debets-Ossenkopp YJ, et al. Prosthetic joint-associated infections treated with DAIR (debridement, antibiotics, irrigation, and retention): analysis of risk factors and local antibiotic carriers in 91 patients. *Acta Orthop*. 2013;84:380–386. doi:10.3109/17453674.2013.823589.
- [13] Kendoff D, Morgan-Jones R, Haddad FS, editors. *Periprosthetic Joint Infections: Changing Paradigms*. Springer International Publishing; 2016.
- [14] Pilge H, Gradl G, von Eisenhart-Rothe R, Gollwitzer H. Incidence and outcome after infection of megaprotheses. *Hip Int*. 2012;S83–S90. doi:10.5301/HIP.2012.9576.
- [15] Brandt CM, Duffy MC, Berbari EF, Hanssen AD, Steckelberg JM, Osmon DR. *Staphylococcus aureus* prosthetic joint infection treated with prosthesis removal and delayed reimplantation arthroplasty. *Mayo Clin Proc*. 1999;74:553–558. doi:10.4065/74.6.553.



Authors: Marjan Wouthuyzen-Bakker, Alex Soriano

QUESTION 9: What factors are associated with the successful treatment of acute periprosthetic joint infection (PJI) using debridement, antibiotics and implant retention (DAIR)?

RECOMMENDATION: The following factors have been shown to be associated with treatment success in acute PJIs treated with DAIR:

- Exchanging the modular components during debridement
- Performing a debridement within at least seven days, but preferably as soon as possible, after the onset of symptoms
- Adding rifampin to the antibiotic regimen, particularly when combined with a fluoroquinolone, in cases of susceptible staphylococci
- Treatment with fluoroquinolones in cases of susceptible gram-negative bacilli

The following factors have been shown to be associated with treatment failure in acute PJIs treated with DAIR:

- Host related factors: rheumatoid arthritis, old age, male sex, chronic renal failure, liver cirrhosis and chronic obstructive pulmonary disease
- Prosthesis indication: fracture as indication for the prosthesis, cemented prostheses and revised prostheses
- Clinical presentation representing the severity of the infection: a high C-reactive protein (CRP), a high bacterial inoculum and the presence of bacteremia
- Causative microorganisms: *S. aureus* and Enterococci

LEVEL OF EVIDENCE: Moderate

DELEGATE VOTE: Agree: 92%, Disagree: 5%, Abstain: 3% (Super Majority, Strong Consensus)

RATIONALE

The success of DAIR depends on multiple host- and implant-related factors, clinical presentation, intraoperative variables, causative microorganism(s) and their antibiotic sensitivities and the antibiotic regimen. It is of note, that the described factors related to treatment outcome in some studies, are not always confirmed by others.

Most factors associated with success of DAIR are demonstrated in retrospective studies, entailing a high risk of selection bias, especially for those factors involving certain treatment strategies. Therefore, prospective validation is critical for most of the described variables and differences between cohorts should be taken into consid-

eration in interpreting risk factors. In addition, the success of DAIR depends on the definition of treatment failure and the total duration of follow-up, which also differed amongst the selected studies.

Factors that are consistently shown in the literature to **increase** the chance of treatment success are:

Exchange of Modular Components

The bacterial load detected on polyethylene is higher compared to metal components of prostheses, presumably due to its rough surface that favors the adherence of bacteria [1]. Therefore, exchanging the modular components will reduce the amount of biofilm present on foreign material. Moreover, removing the modular components during DAIR (i.e., femoral head and/or polyethylene component) provides better access to the joint capsule for radical debridement. Tsang et al. reviewed all cohort studies published between 1977 and 2015 on the outcome of DAIR in hip PJI. The success rate of DAIR in studies where all patients underwent modular component exchange was 73.9% (471/637 patients; 95% confidence interval (CI), 70 to 77) compared to 60.7% (245/404 patients; 95% CI, 56 to 65) in patients in whom modular components were retained ($p < 0.0001$) [2]. In addition, Grammatopoulos et al. demonstrated in a cohort of 82 acute hip PJIs a treatment success of 93.3% when modular components were exchanged versus 75.7% when modular component were retained ($p = 0.02$) [3]. Smaller studies confirm the same in acute PJIs of the knee [4,5]. The beneficial effect of modular exchange was also demonstrated as independent predictors of treatment success in large multi-center cohort studies evaluating the outcome of DAIR in hip and knee PJIs caused by methicillin-resistant and methicillin susceptible *S.aureus* ($n = 345$, hazard ratio (HR) 0.65, $p < 0.026$) [6], streptococci ($n = 462$, HR 0.60, $p < 0.01$) [7] and solely late acute PJIs ($n = 340$, odds ratio (OR) 0.35, $p = 0.02$).

Performing DAIR within at Least Seven Days after the Onset of Symptoms

Several studies demonstrated that the duration of symptoms are significantly shorter in patients who were successfully treated with DAIR compared to patients in whom treatment failed [8–13]. In most studies, the most prominent difference between success and failure is observed using a symptom duration of one week as optimal cut-off [3,10,11,14,15]. Urish et al. demonstrated a treatment success rate of 53.2% in 216 knee PJIs when DAIR was performed within one week after the onset of symptoms. Additional multivariate analysis in this study showed that the chance of failure increased when DAIR was postponed to two weeks after onset of symptoms (HR 1.68), and further increased after four weeks of symptoms (HR 2.34) ($p = 0.002$) [14]. Grammatopoulos et al. demonstrated a treatment success rate of 90.7% in 82 hip PJIs when DAIR was performed within one week after the onset of symptoms versus 75.0% when DAIR was performed after one week ($p = 0.05$) [3]. As the maximum days of symptom duration was not well described in all studies and chronic PJIs are indeed included in some [3,10,12,14], the beneficial effect of debridement within one week may be overestimated in these studies for solely acute PJIs. However, a study performed in 110 patients who had a maximum of 32 days of symptoms indicates the same conclusion [8,9]. These authors demonstrated that for each additional day of postponing DAIR, the odds of implant retention decreased by 15.7% and 7.5% for hip and knee PJI, respectively. In the same study, multivariate analysis showed that performing a DAIR within five days was an independent predictor for treatment success, with an OR of around 0.05 for both hips and knees (95% CI 0.01 to 0.24). These data support the concept that a DAIR should be performed within one week to increase the chance of treatment success, but should preferably be performed as soon as possible.

The Addition of Rifampin in Staphylococci PJI

In the randomized controlled trial performed by Zimmerli et al. in 1998, 24 patients with an infected orthopaedic implant caused by staphylococci and treated with surgical debridement were randomized to antimicrobial treatment with combination ciprofloxacin/rifampin or with ciprofloxacin monotherapy. Adding rifampin to the antibiotic regimen improved treatment success from 58 - 100% ($p = 0.02$) [16]. Although relatively small in sample size, this study served as the foundation of adding rifampin to the antibiotic regimen in staphylococcal PJI. Thereafter, the benefit of rifampin was primarily demonstrated in observational studies [6,17–19]. In a prospective study including 86 monomicrobial staphylococci knee PJIs treated with open debridement, rifampin-based regimens had a 40% higher treatment success compared to other regimens ($p = 0.01$) [17]. Moreover, the addition of rifampin has shown to be a strong independent predictor for treatment success in multivariate analyses [6,20]. The greatest beneficial effect of rifampin has been shown when combined with a fluoroquinolone, which can be explained by the effectivity of fluoroquinolones against biofilm and by drug-interactions of rifampin with several other antibiotics but not with levofloxacin, the most frequently used fluoroquinolone. In a retrospective study of gram-positive infections treated with DAIR, Tornero et al. demonstrated that rifampin combined with linezolid, co-trimoxazole or clindamycin (which are known to have a drug-interaction with rifampin) was associated with a higher failure rate (27.8%) compared to a combination of rifampin with levofloxacin, ciprofloxacin or amoxicillin (8.3%) ($p = 0.026$) [19]. The greater benefit of the fluoroquinolone-rifampin combination therapy compared to other antibiotic regimens was also illustrated by Puhto et al. in a study of 113 patients with acute PJI: compared to rifampin-ciprofloxacin, the HR for treatment failure was significantly increased in the rifampin-other antibiotics group (HR 6.0, 95% CI 1.5 to 28.8, $p = 0.014$), and even higher in patients treated without rifampin (HR 14.4, 95% CI 3.1 to 66.9, $p < 0.01$) [20]. In addition, Senneville et al., observed the same in 41 patients with acute *S. aureus* PJI treated with DAIR: treatment success was 93.8% in the fluoroquinolone-rifampin group, 66.7% in the rifampin-other antibiotics group and 57.1% in regimens without rifampin ($p = 0.11$) [21]. Altogether, these data indicate that adding rifampin to the antibiotic regimen, particularly when combined with a fluoroquinolone, is associated with an increased chance of treatment success in acute PJI treated with DAIR.

The Use of Fluoroquinolones in Gram-negative PJI

The protective effect of antibiotic treatment with a fluoroquinolone is demonstrated in two prospective and one retrospective observational study [19,22,23]. In a prospective cohort of 22 patients with early PJI caused by gram-negative organisms, the use of fluoroquinolones was associated with a lower failure rate (7.1%) compared to other antibiotic regimens (37.5%) ($p = 0.04$) [19]. In addition, in a cohort study of 47 cases, treatment with a fluoroquinolone in susceptible gram-negative bacilli was associated with a better outcome ($p = 0.0009$) and was an independent predictor of treatment success (OR, 9.09; 95% CI, 1.96 to 50; $p = 0.005$) [23]. Finally, a large retrospective, multicenter study on gram-negative PJI was performed in 16 Spanish hospitals in which DAIR was performed in 72% of the cases (174/242 cases) [22]. The overall success rate of DAIR was 68%, which increased to 79% in gram-negative PJIs treated with ciprofloxacin. In agreement with the previous study, ciprofloxacin treatment exhibited an independent protective effect in the multivariate analysis (HR 0.23; 95% CI, 0.13 to 0.40; $p < 0.001$). In all of these studies, no propensity score matching was performed to correct for possible selection bias. In addition, it should be noted that in most of the performed studies, oral therapy with fluoroquinolones was compared with oral beta-

lactam antibiotics. Questioning the superiority of fluoroquinolones, Grossi et al. demonstrated that treatment with high dose intravenous beta-lactam antibiotics (alone or with the addition of another antimicrobial agent) was not inferior to treatment with fluoroquinolones [24]. Although this study had a relatively small sample size ($n = 76$) and included both DAIRs and staged revision surgeries, it does provide some evidence for the possibility that alternative intravenous antibiotic regimens and/or combination therapy may be as effective as treatment with fluoroquinolones. More studies are required to confirm this finding.

Factors that are consistently shown in the literature to **decrease** the chance of treatment success are:

Host-related Factors

The importance of host factors in the outcome of patients with a PJI was highlighted by McPherson et al., who described the first grading of the medical and immune status of the host to predict outcome [25]. However, this grading system was not validated in large cohorts of patients who underwent DAIR. For patients managed with DAIR, three large cohort studies in streptococci, staphylococci and late acute PJI identified patients with rheumatoid arthritis (RA) as an important risk factor for failure [6,7]. This high risk for failure in RA patients has been demonstrated in smaller studies as well [10,26,27]. The most pronounced risk was observed for late acute PJIs, demonstrating a failure rate of 74% in patients with RA versus 43% in patients without ($p < 0.001$), and was shown to be an independent predictor for failure in the multivariate analysis, with an OR of 5.1 (95% CI 1.1 – 24.3, $p = 0.04$). Age has been independently associated with worse outcome in a recent large cohort of late acute PJIs, showing that patients older than 80 years old had a significantly higher risk of failure (OR 2.6). In addition, a clear correlation between treatment failure and age has also been described in a large cohort of early PJIs [28]. Male sex [28], chronic renal failure [7,22,29] and liver cirrhosis [29,30] were also identified as independent predictors of failure in patients treated with DAIR. Patients with chronic obstructive pulmonary disease (COPD) showed an increased risk for failure in late acute PJIs only. In this study, COPD was not a significant predictor for failure in the multivariate analysis (OR 2.9, 95% CI 0.99 – 8.68, $p < 0.05$).

Prosthesis Indication

Despite the fact that fracture and revision arthroplasties have a higher predisposition for infection [31–34], these arthroplasties have been associated with a higher risk for treatment failure in acute PJIs as well. Fracture as an indication for the prosthesis has been shown to be associated with DAIR failure in three studies of early acute PJIs [28,29,35] and in one study of late acute PJIs as well. With an average failure rate that is 20–30% higher compared to osteoarthritis, fracture as an indication for prosthesis has been shown to be an independent predictor for treatment failure in two studies [29]. The same holds true for revision arthroplasty compared to infected primary arthroplasty, with a failure rate that is 12–22% higher [29,36], and even higher in knees [4]. Revision arthroplasty has been shown to be an independent predictor for failure in early acute PJI [29,36]. Only one study demonstrated an increased risk for failure in cemented prostheses, with an OR of 8.7 in the multivariate analysis [29].

Clinical Presentation

Several factors considered as surrogate parameters for the severity of the infection have been associated with treatment failure: a high CRP at clinical presentation [6,23,28,29,37], the amount/percentage of positive intraoperative cultures representing the bacterial inoculum [28,29] and bacteremia/sepsis [7,28,29,38]. In most

of these studies, these factors are closely correlated to one another. In case of CRP value, an average cut-off value of > 115 mg/L has been associated with an increased failure rate, depending on the type of infection (late acute or early acute). Notably, late acute/hematogenous infections appear to be associated with worse outcomes compared to early acute/post-surgical infections, especially when the infection is caused by *S. aureus* [6,15,20,37–41].

Causative Microorganism

It has been demonstrated in several studies that an infection caused by *S. aureus* is associated with an increased risk of failure [28,36,42,43]. In a large retrospective cohort of 386 early acute PJIs performed by Löwik et al., the percentage of failure was 17% higher when the infection was caused by *S. aureus* compared to other microorganisms (47.5% vs. 30.2%, $p < 0.001$). *S. aureus* infection was also a prominent risk factor for failure in late acute PJIs, illustrated by an OR of 3.52 for *S. aureus* in the multivariate analysis. Methicillin-resistant *S. aureus* (MRSA) infection was associated with an increased risk for failure in a study performed by Cobo et al., but this was not demonstrated as an independent variable in the multivariate analysis [40]. Indeed, Lora-Tamayo et al. clearly demonstrated that MRSA infections have similar failure rates as methicillin-susceptible *S. aureus*, although the time to failure differs [6]. Next to *S. aureus*, overall, poor outcomes have been described for enterococcal PJIs [43–46]. The largest analysis on enterococcal PJI have been performed by Tornero et al., who reported a failure rate of 53% in 94 patients treated with DAIR [45]. Subanalysis demonstrated that infection caused by *E. faecium* have a worse outcome than those caused by *E. faecalis* (72% vs. 42% failure, $p < 0.04$). Indeed, two studies identified the presence of enterococci as an independent risk factor for failure in acute PJI treated with DAIR [43].

Ultimately, a clinical risk score including the most potent factors associated with treatment failure and treatment success should be developed to predict the individual chance of treatment success. One of the main objectives of risk scores would be to identify patients with high failure rate using DAIR. To be of most clinical use, these scores should preferably include preoperative variables only. So far, two articles described a risk score for failure in early acute PJIs (KLIC-score, Fig. 1A) [29] and late acute PJIs (CRIME80-score, Fig. 1B) treated with DAIR. These risk scores can aid in the clinical decision making to choose an alternative surgical approach and/or to intensify the antimicrobial regimen.

REFERENCES

- [1] Lass R, Giurea A, Kubista B, Hirschl AM, Graninger W, Prestler E, et al. Bacterial adherence to different components of total hip prosthesis in patients with prosthetic joint infection. *Int Orthop*. 2014;38:1597–602. doi:10.1007/s00264-014-2358-2.
- [2] Tsang S-TJ, Ting J, Simpson AHRW, Gaston P. Outcomes following debridement, antibiotics and implant retention in the management of periprosthetic infections of the hip: a review of cohort studies. *Bone Joint J*. 2017;99-B:1458–66. doi:10.1302/0301-620X.99B11.BJJ-2017-0088.R1.
- [3] Grammatopoulos G, Bolduc M-E, Atkins BL, Kendrick BJL, McLardy-Smith P, Murray DW, et al. Functional outcome of debridement, antibiotics and implant retention in periprosthetic joint infection involving the hip: a case-control study. *Bone Joint J*. 2017;99-B:614–22. doi:10.1302/0301-620X.99B5.BJJ-2016-0562.R2.
- [4] Zhang C, Yan CH, Chan PK, Ng FY, Chiu KY. Polyethylene insert exchange is crucial in debridement for acute periprosthetic infections following total knee arthroplasty. *J Knee Surg*. 2017;30:36–41. doi:10.1055/s-0036-1579667.
- [5] Choi H-R, von Knoch F, Zurakowski D, Nelson SB, Malchau H. Can implant retention be recommended for treatment of infected TKA? *Clin Orthop Relat Res*. 2011;469:961–969. doi:10.1007/s11999-010-1679-8.
- [6] Lora-Tamayo J, Murillo O, Iribarren JA, Soriano A, Sánchez-Somolinos M, Baraia-Etxaburu JM, et al. A large multicenter study of methicillin-susceptible and methicillin-resistant *Staphylococcus aureus* prosthetic joint infections managed with implant retention. *Clin Infect Dis*. 2013;56:182–194. doi:10.1093/cid/cis746.
- [7] Lora-Tamayo J, Senneville É, Ribera A, Bernard L, Dupon M, Zeller V, et al. The not-so-good prognosis of streptococcal periprosthetic joint infection

TABLE 1. Literature review of factors associated with successful treatment of acute PJI using debridement, antibiotics, and implant retention

Author, Year	N	PJI	Variables	Failure Rate	Univariate (OR or HR) ⁷	Multivariate (OR or (a)HR) ⁷
Tsang, 2017 [2] Meta-analysis	1296	Early & late	Symptoms ≤ 7 d vs. > 7 d Exchange of modular components (yes vs. no)	28% vs. 48%, p = 0.0001 26% vs. 39%, p = 0.0001	-	-
Grammatopoulos, 2017 [3]	82	Early & late	Symptoms ≤ 7 d vs. > 7 d Interval since arthroplasty ≤ 6 w vs. > 6 w Exchange of modular components (yes vs. no)	9% vs. 25%, p = 0.05 7.5% vs. 27.5%, p = 0.01 6.6% vs. 24.4%, p = 0.02	-	-
Zhang, 2017 [4]	34	Early & late	Exchange of modular components (yes vs. no)	39% vs. 100%, p = 0.008	-	-
Choi, 2011 [5]	32	Early & late	Exchange of modular components (yes vs. no)	47% vs. 100%, p = 0.001	-	-
Lora-Tamayo, 2013 [6]	345	Early & late	Immunosuppression (yes vs. no) Bacteremia (yes vs. no) Polymicrobial (yes vs. no) CRP Exchange of modular components (yes vs. no) Need of ≥ 2 debridements (yes vs. no) ¹ levofloxacin+rifampin ³ vancomycin+rifampin	71% vs. 43%, p = 0.006 65% vs. 41%, p = 0.001 59% vs. 41%, p = 0.005 NP, p = 0.001 41% vs. 56%, p = 0.004 71% vs. 41%, p = 0.003 NP, p = 0.008 NP, p = 0.02	2.31 2.29 1.76 1.29 0.56 1.98 0.50 0.34	2.23 1.81 1.77 1.22 0.65 1.63 0.42 0.29
Lora-Tamayo, 2017 [7] ⁸	462	Early & late	⁸ Chronic renal failure (yes vs. no) ⁸ Rheumatoid arthritis (yes vs. no) ⁸ Immunosuppression (yes vs. no) ⁸ Revision (yes vs. no) ⁸ Late post-surgical infection (yes vs. no) ⁸ Bacteremia (yes vs. no) ⁸ Exchange of modular components (yes vs. no)	54.5% vs. 40.8%, p = 0.05 64.9% vs. 40.0%, p < 0.01 60.4% vs. 39.9%, p < 0.01 53.6% vs. 38.3%, p < 0.01 62.9% vs. 38.2%, p < 0.01 47.7% vs. 37.9%, p = 0.02 33.0% vs. 51.6%, p < 0.01	1.58 2.23 1.86 1.60 1.41 1.44 0.59	- 2.36 - 1.37 2.20 1.69 0.60
Wouthuyzen-Bakker, 2018 [8]	340	Late	Gender, male vs. female Age, > 80 y vs. ≤ 80 y old COPD (yes vs. no) Active malignancy (yes vs. no) RA (yes vs. no) Immunosuppression Immunosuppression (yes vs. no) Fracture (yes vs. no) Revision (yes vs. no) CRP > 150 vs. ≤ 150 mg/L Bacteremia (yes vs. no) S. aureus (yes vs. no) Exchange of modular components (yes vs. no)	49.1% vs. 40.6%, p = 0.11 54.8% vs. 42.3%, p = 0.06 55.9% vs. 43.8%, p = 0.18 51.7% vs. 44.4%, p = 0.04 74.1% vs. 42.5%, p = 0.001 61.5% vs. 42.9%, p = 0.03 70.6% vs. 41.9%, p = 0.02 54.2% vs. 41.7%, p = 0.04 47.9% vs. 41.7%, p = 0.06 56% vs. 39.8%, p = 0.005 53.9% vs. 38.7%, p = 0.005 36.4% vs. 52.4%, p = 0.004	-	2.02 2.60 2.90 - 5.13 - 5.39 - 2.00 - 3.52 0.35
Urish, 2017 [14]	206	Early & late	Symptoms ≤ 7 d vs. > 7 d S. aureus vs. other	NP, p = 0.004 NP, p = 0.04	1.77 0.63	1.68 0.59
Koh, 2015 [15]	52	Early & late	Early vs. late PJI	18.7% vs. 47.3%, p = 0.04	-	-

TABLE 1. Literature review of factors associated with successful treatment of acute PJI using debridement, antibiotics, and implant retention (Cont.)

Author, Year	N	PJI	Variables	Failure Rate	Univariate (OR or HR) ⁷	Multivariate (OR or (a)HR) ⁷
Triantafyllopoulos, 2015 [9]	78	NP	Thyroid disease Duration of symptoms MR-staphylococci	68.7%, $p = 0.03$ $p = 0.0001$ 57%, $p = 0.004$	-	-
Kuiper, 2013 [10]	91	Early & late	RA (yes vs. no) Symptoms ≤ 7 d vs. > 7 d Early vs. late PJI ESR > 60 mm/h CNS vs. others	70% vs. 30%, $p = 0.03$ 26.6% vs. 48.4%, $p = 0.02$ 31% vs. 71.4%, $p = 0.04$ NP, $p = 0.001$ 69% vs. 28%, $p = 0.009$	-	1.2-8.4 ¹ 1-18 ¹ 1.1-366 ¹ 2.2-98 ¹ 1.8-309 ¹
Marculescu, 2006 [11]	99	Early & late	Sinus tract Symptoms > 8 d	61%, $p = 0.002$ 51%, $p = 0.04$	2.85 1.79	2.84 1.77
Buller, 2012 [12]	309	Early & late	Symptoms < 21 d vs. ≥ 21 d ESR Previous infection in the same joint (yes vs. no) Resistant-GP vs. others	NP, $p = 0.001$ $p = 0.02$ 55% vs. 44%, $p = 0.009$ 65% vs. 44%, $p = 0.005$	-	-
Hsieh, 2009 [13]	154	Early & late	GN vs. GP	73% vs. 53%, $p = 0.002$	-	-
Tornero, 2016 [16]	143	Early	Suboptimal vs. optimal (rifampin for GP and FQ for GN) antibiotic treatment	31% vs. 8%, $p = 0.004$	-	4.92
Puhto, 2015 [20]	113	Early & late	Early vs. late PJI Leukocytes $>$ vs. $\leq 10 \times 10^9/L$ Ineffective empirical antibiotics vs. effective ⁴ Rifampin+ciprofloxacin vs. Rifampin+other vs. other	30.8% vs. 54.3%, $p = 0.002$ 50% vs. 24.6%, $p < 0.01$ 60% vs. 33%, $p < 0.006$ 10% vs. 40% vs. 70%, $p < 0.01$	- R+C vs. R+O: 6 R+C vs. O: 14	- 3.7 3.2 -
Holmberg, 2015 [17]	145	Early & late	Revision (yes vs. no) Rifampin vs. no rifampin	63% vs. 23%, $p = 0.02$ 19% vs. 59%, $p = 0.01$	-	-
Vilchez, 2011 [38]	65	Early & late	Early vs. late PJI Need of ≥ 2 debridements	24.5% vs. 58.7%, $p = 0.02$ NP, $p = 0.001$	-	2.57 4.61
El Helou, 2010 [18]	91	Early & late	Rifampin vs. no rifampin	4% vs. 40%, $p = 0.03$	-	0.11
Zimmerli, 1998 [16] ⁵	18	Early	Rifampin+ciprofloxacin vs. ciprofloxacin	100% vs. 58%, $p = 0.02$	-	-
Senneville, 2011 [21]	41	Early & late	Rifampin+FQ vs. other	6% vs. 32%, $p = 0.001$	-	-
Martínez-Pastor, 2009 [23]	47	Early & late	FQ vs. no FQ for GN PJI CRP $>$ vs. ≤ 15 mg/dL	7% vs. 52%, $p = 0.005$ 50% vs. 17%, $p = 0.04$	-	9.09 3.57
Tornero, 2015 [29]	222	Early	Chronic renal failure (yes vs. no) Liver cirrhosis (yes vs. no) Femoral neck fracture / revision surgery vs. primary Cemented prosthesis (yes vs. no) CRP $>$ vs. ≤ 11.5 mg/dL	60% vs. 20%, $p < 0.001$ 48% vs. 21%, $p = 0.004$ 35% / 38% vs. 16%, $p = 0.003$ 25% vs. 19%, $p = 0.39$ 56% vs. 16%, $p < 0.001$	-	5.92 4.46 4.39 / 4.34 8.71 12.3
Rodríguez-Pardo, 2014 [22]	174	Early & late	Ciprofloxacin (yes vs. no) Chronic renal failure	21% vs. 60%, $p < 0.001$ NP, $p < 0.02$	-	0.23 2.56
Grossi, 2016 [24]	35	Early & late	Ciprofloxacin (yes vs. no)	21% vs. 28%, $p = 0.65$	-	-

TABLE 1. Literature review of factors associated with successful treatment of acute PJI using debridement, antibiotics, and implant retention (Cont.)

Author, Year	N	PJI	Variables	Failure Rate	Univariate (OR or HR) ⁷	Multivariate (OR or (a)HR) ⁷
Löwik, 2018 [28]	386	Early	CRP >115 vs. ≤115 mg/L Gender, male vs. female Left-sided prosthesis (yes vs. no) Sepsis (yes vs. no) Ischaemic heart disease (yes vs. no) Fracture (yes vs. no) Gentamicin impregnated beads or sponges (yes vs. no) S. aureus (yes vs. no)	55.2% vs. 30.3%, $p < 0.001$ 46.6% vs. 33.2%, $p = 0.08$ 46.7% vs. 31.1%, $p = 0.002$ 52.1% vs. 35.1%, $p = 0.007$ 50.6% vs. 35.3%, $p = 0.013$ 52.8% vs. 33.3%, $p = 0.047$ 43.0% vs. 23.7%, $p = 0.001$ 50.2% vs. 36.6%, $p = 0.022$	-	- 2.03 1.80 - 1.84 - NP NP
Hsieh, 2013 [26]	154	Early & late	RA (yes vs. no)	78% vs. 48%, $p = 0.002$	-	-
Son, 2017 [27]	25	Early & late	RA (yes vs. no)	50% vs. 5%, $p = 0.04$	-	-
Tornero, 2014 [30]	160	Early	Liver cirrhosis (yes vs. no) CRP > vs. ≤12 mg/dL GN not treated with a FQ vs. treated with a FQ	67% vs. 29%, $p < 0.001$ 47% vs. 29%, $p = 0.04$ 57% vs. 31.5%, $p = 0.005$	-	12.4 1.06 6.5
Bergkvist, 2016 [35]	35	Early	Hip fracture (yes vs. no)	64% vs. 19%, $p = 0.01$	-	8.3
Byren, 2009 [36]	112	Early & late	Arthroscopy vs. open S. aureus vs. others Revision vs. primary	53% vs. 12%, $p = 0.008$ 30% vs. 24%, $p = 0.05$ 34.6% vs. 12.8%, $p = 0.008$	5.4 2.6 2.6	4.2 2.9 3.1
Vilchez, 2011 [37]	53	Early	CRP > vs. ≤ 22 mg/dL Need of 2 nd debridement (yes vs. no)	54.5% vs. 16.6%, $p = 0.01$ 75% vs. 18.4%, $p = 0.006$	-	20.4 9.8
Rodriguez, 2010 [39]	50	Late	S. aureus GN	62.5%, $p = 0.01$ 0%, $p = 0.01$	3.08 0.46	5.3 0.6
Cobo, 2011 [40]	139	Early	MRSA (yes vs. no)	66.6% vs. 39.6%, $p = 0.05$	-	None
Tande, 2016 [41]	43	Late		66.6% vs. 39.6%, $p = 0.05$		
Letouvet, 2016 [42]	60	Early & Late	Number of prior surgeries S. aureus (yes vs. no) Antibiotic treatment < 3 months	$p = 0.03$ 50% vs. 22%, $p = 0.02$ 46% vs. 23.5%, $p = 0.01$	2.7 3.4	6.3 9.4 20
Soriano, 2006 [43]	47	Early	Enterococcus spp or MRSA vs. others	87.5% vs. 9%, $p = 0.003$	-	17.6
Kheir, 2017 [44] ⁶	87	Early & Late	VSE VRE Polymicrobial with enterococci	35% 50% 56%	-	-
Tornero, 2014 [45] ⁶	203	Early & Late	VSE VRE	41.8% 72%	-	-
Duijf, 2015 [46]	44	Early	Enterococcus sp	34%	-	-

CRP, C-reactive protein; PJI, periprosthetic joint infection; NP, information not provided; MR, methicillin-resistant; ESR, erythrocyte-sedimentation rate; CNS, coagulase-negative staphylococci; GP, gram-positive cocci; GN, gram-negative bacilli; FQ, fluoroquinolone; VSE, vancomycin-susceptible enterococci; VRE, vancomycin-resistant enterococci; RA, rheumatoid arthritis.

¹ Confidence interval 95%.

² Sub-group analysis of patients with a post-surgical PJI due to methicillin-susceptible *S. aureus* (MSSA).

³ Sub-group analysis of patients with a post-surgical PJI due to methicillin-resistant *S. aureus* (MRSA).

⁴ Sub-group analysis of patients with a post-surgical PJI due to staphylococci.

⁵ Randomized, placebo-controlled, double-blind trial.

⁶ Including patients treated with DAIR and prosthesis exchange.

⁷ Only depicted when p -value < 0.05.

⁸ Only depicting the results associated with overall failure.

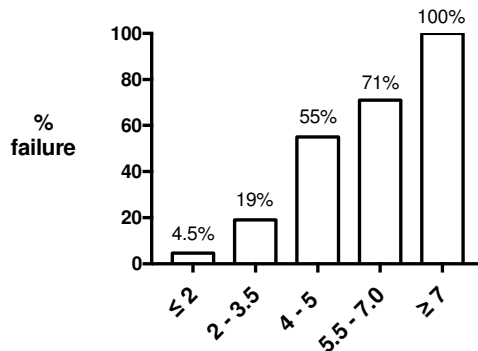


FIGURE 1A. KLIC preoperative risk score [19,28]

K	Chronic renal failure (Kidney)	2
L	Liver cirrhosis	1.5
I	Index surgery:	
	indication prosthesis: fracture OR	1.5
	revision prosthesis	
C	Cemented prosthesis	2
	CRP > 115 mg/L	2.5

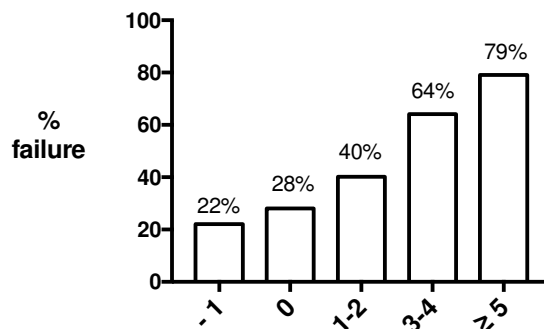


FIGURE 1B. CRIME80 preoperative risk score [19,28]

C	COPD	2
	CRP > 150 mg/L	1
R	Rheumatoid arthritis	3
I	Indication prosthesis: fracture	3
M	Male	1
E	Exchange of mobile components	-1
80	Age > 80 years	2

managed by implant retention: the results of a large multicenter study. Clin Infect Dis. 2017;64:1742-1752.

- [8] Triantafyllopoulos GK, Poultsides LA, Sakellariou VI, Zhang W, Sculco PK, Ma Y, et al. Irrigation and debridement for periprosthetic infections of the hip and factors determining outcome. Int Orthop. 2015;39:1203-1209. doi:10.1007/s00264-015-2753-3.
- [9] Triantafyllopoulos GK, Poultsides LA, Zhang W, Sculco PK, Ma Y, Sculco TP. Periprosthetic knee infections treated with irrigation and debridement: outcomes and preoperative predictive factors. J Arthroplasty. 2015;30:649-657. doi:10.1016/j.arth.2014.10.026.
- [10] Kuiper JWP, Vos SJC, Saouti R, Vergroesen DA, Graat HCA, Debets-Ossenkopp YJ, et al. Prosthetic joint-associated infections treated with DAIR (debridement, antibiotics, irrigation, and retention): analysis of risk factors and local antibiotic carriers in 91 patients. Acta Orthop. 2013;84:380-386. doi:10.3109/17453674.2013.823589.
- [11] Marculescu CE, Berbari EF, Hanssen AD, Steckelberg JM, Harmsen SW, Mandrekar JN, et al. Outcome of prosthetic joint infections treated with debridement and retention of components. Clin Infect Dis. 2006;42:471-478. doi:10.1086/499234.
- [12] Buller LT, Sabry FY, Easton RW, Klika AK, Barsoum WK. The preoperative prediction of success following irrigation and debridement with polyethylene exchange for hip and knee prosthetic joint infections. J Arthroplasty. 2012;27:857-864.e1-4. doi:10.1016/j.arth.2012.01.003.
- [13] Hsieh PH, Lee MS, Hsu KY, Chang YH, Shih HN, Ueng SW. Gram-negative prosthetic joint infections: risk factors and outcome of treatment. Clin Infect Dis. 2009;49:1036-1043. doi:10.1086/605593.
- [14] Urish KL, Bullock AG, Kreger AM, Shah NB, Jeong K, Rothenberger SD, et al. A multicenter study of irrigation and debridement in total knee arthroplasty periprosthetic joint infection: treatment failure is high. J Arthroplasty. 2017.
- [15] Koh JJ, Han SB, In Y, Oh KJ, Lee DH, Kim TK, et al. Open debridement and prosthesis retention is a viable treatment option for acute periprosthetic joint infection after total knee arthroplasty. Arch Orthop Trauma Surg. 2015;135:847-855. doi:10.1007/s00402-015-2237-3.
- [16] Zimmerli W, Widmer AF, Blatter M, Frei R, Ochsner PE. Role of rifampin for treatment of orthopedic implant-related staphylococcal infections: a randomized controlled trial. Foreign-Body Infection (FBI) Study Group. JAMA. 1998;279:1537-1541.
- [17] Holmberg A, Thórhallsdóttir VG, Robertsson O, W-Dahl A, Stefánsdóttir A. 75% success rate after open debridement, exchange of tibial insert, and antibiotics in knee prosthetic joint infections. Acta Orthop. 2015;86:457-462. doi:10.3109/17453674.2015.1026756.
- [18] El Helou OC, Berbari EF, Lahr BD, Eckel-Passow JE, Razonable RR, Sia IG, et al. Efficacy and safety of rifampin containing regimen for staphylococcal prosthetic joint infections treated with debridement and retention. Eur J Clin Microbiol Infect Dis. 2010;29:961-967. doi:10.1007/s10096-010-0952-9.
- [19] Tornero E, Morata L, Martínez-Pastor JC, Angulo S, Combalá A, Borí G, et al. Importance of selection and duration of antibiotic regimen in prosthetic

- joint infections treated with debridement and implant retention. *J Antimicrob Chemother.* 2016;71:1395–1401. doi:10.1093/jac/dkv481.
- [20] Puhto AP, Puhto T, Niinimäki T, Ohtonen P, Leppilähti J, Syrjälä H. Predictors of treatment outcome in prosthetic joint infections treated with prosthesis retention. *Int Orthop.* 2015;39:1785–1791. doi:10.1007/s00264-015-2819-2.
- [21] Senneville E, Joulie D, Legout L, Valette M, Dezéque H, Beltrand E, et al. Outcome and predictors of treatment failure in total hip/knee prosthetic joint infections due to *Staphylococcus aureus*. *Clin Infect Dis.* 2011;53:334–340. doi:10.1093/cid/cir402.
- [22] Rodríguez-Pardo D, Pigrau C, Lora-Tamayo J, Soriano A, del Toro MD, Cobo J, et al. Gram-negative prosthetic joint infection: outcome of a debridement, antibiotics and implant retention approach. A large multicentre study. *Clin Microbiol Infect.* 2014;20:0911–0919. doi:10.1111/1469-0691.12649.
- [23] Martínez-Pastor JC, Muñoz-Mahamud E, Vilchez F, García-Ramiro S, Bori G, Sierra J, et al. Outcome of acute prosthetic joint infections due to gram-negative bacilli treated with open debridement and retention of the prosthesis. *Antimicrob Agents Chemother.* 2009;53:4772–4777. doi:10.1128/AAC.00188-09.
- [24] Grossi O, Asseray N, Bourigault C, Corvec S, Valette M, Navas D, et al. Gram-negative prosthetic joint infections managed according to a multidisciplinary standardized approach: risk factors for failure and outcome with and without fluoroquinolones. *J Antimicrob Chemother.* 2016;71:2593–2597. doi:10.1093/jac/dkw202.
- [25] McPherson EJ, Woodson C, Holtom P, Roidis N, Shufelt C, Patzakis M. Periprosthetic total hip infection: outcomes using a staging system. *Clin Orthop Relat Res.* 2002;8–15.
- [26] Hsieh PH, Huang KC, Shih HN. Prosthetic joint infection in patients with rheumatoid arthritis: an outcome analysis compared with controls. *PLoS ONE.* 2013;8:e71666. doi:10.1371/journal.pone.0071666.
- [27] Son WS, Shon OJ, Lee DC, Park SJ, Yang HS. Efficacy of open debridement and polyethylene exchange in strictly selected patients with infection after total knee arthroplasty. *Knee Surg Relat Res.* 2017;29:172–179.
- [28] Löwik CAM, Jutte PC, Tornero E, Ploegmakers JJW, Knoben BAS, de Vries AJ, et al. Predicting failure in early acute prosthetic joint infection treated with debridement, antibiotics, and implant retention: external validation of the KLIC Score. *J Arthroplasty.* 2018.
- [29] Tornero E, Morata L, Martínez-Pastor JC, Bori G, Climent C, García-Velez DM, et al. KLIC-score for predicting early failure in prosthetic joint infections treated with debridement, implant retention and antibiotics. *Clin Microbiol Infect.* 2015;21:786.e9–786.e17. doi:10.1016/j.cmi.2015.04.012.
- [30] Tornero E, Martínez-Pastor JC, Bori G, García-Ramiro S, Morata L, Bosch J, et al. Risk factors for failure in early prosthetic joint infection treated with debridement. Influence of etiology and antibiotic treatment. *J Appl Biomater Funct Mater.* 2014;12:129–134. doi:10.5301/jabfm.5000209.
- [31] Guren E, Figved W, Frihagen F, Watne LO, Westberg M. Prosthetic joint infection—a devastating complication of hemiarthroplasty for hip fracture. *Acta Orthop.* 2017;88:383–389. doi:10.1080/17453674.2017.1301009.
- [32] Kurtz SM, Lau E, Schmier J, Ong KL, Zhao K, Parvizi J. Infection burden for hip and knee arthroplasty in the United States. *J Arthroplasty.* 2008;23:984–991. doi:10.1016/j.arth.2007.10.017.
- [33] Mortazavi SMJ, Molligan J, Austin MS, Purtill JJ, Hozack WJ, Parvizi J. Failure following revision total knee arthroplasty: infection is the major cause. *Int Orthop.* 2011;35:1157–1164. doi:10.1007/s00264-010-1134-1.
- [34] Mortazavi SMJ, Schwartzberger J, Austin MS, Purtill JJ, Parvizi J. Revision total knee arthroplasty infection: incidence and predictors. *Clin Orthop Relat Res.* 2010;468:2052–2059. doi:10.1007/s11999-010-1308-6.
- [35] Bergkvist M, Mukka SS, Johansson L, Ahl TE, Sayed-Noor AS, Sköldenberg OG, et al. Debridement, antibiotics and implant retention in early periprosthetic joint infection. *Hip Int.* 2016;26:138–143. doi:10.5301/hipint.5000328.
- [36] Byren I, Bejon P, Atkins BL, Angus B, Masters S, McLardy-Smith P, et al. One hundred and twelve infected arthroplasties treated with “DAIR” (debridement, antibiotics and implant retention): antibiotic duration and outcome. *J Antimicrob Chemother.* 2009;63:1264–1271. doi:10.1093/jac/dkp107.
- [37] Vilchez F, Martínez-Pastor JC, García-Ramiro S, Bori G, Maculé F, Sierra J, et al. Outcome and predictors of treatment failure in early post-surgical prosthetic joint infections due to *Staphylococcus aureus* treated with debridement. *Clin Microbiol Infect.* 2011;17:439–444. doi:10.1111/j.1469-0691.2010.03244.x.
- [38] Vilchez F, Martínez-Pastor JC, García-Ramiro S, Bori G, Tornero E, García E, et al. Efficacy of debridement in hematogenous and early post-surgical prosthetic joint infections. *Int J Artif Organs.* 2011;34:863–869. doi:10.5301/ijao.5000029.
- [39] Rodríguez D, Pigrau C, Euba G, Cobo J, García-Lechuz J, Palomino J, et al. Acute hematogenous prosthetic joint infection: prospective evaluation of medical and surgical management. *Clin Microbiol Infect.* 2010;16:1789–1795. doi:10.1111/j.1469-0691.2010.03157.x.
- [40] Cobo J, Miguel LGS, Euba G, Rodríguez D, García-Lechuz JM, Riera M, et al. Early prosthetic joint infection: outcomes with debridement and implant retention followed by antibiotic therapy. *Clin Microbiol Infect.* 2011;17:1632–1637. doi:10.1111/j.1469-0691.2010.03333.x.
- [41] Tande AJ, Palraj BR, Osmon DR, Berbari EF, Baddour LM, Lohse CM, et al. Clinical presentation, risk factors, and outcomes of hematogenous prosthetic joint infection in patients with *Staphylococcus aureus* Bacteremia. *Am J Med.* 2016;129:221.e11–e20. doi:10.1016/j.amjmed.2015.09.006.
- [42] Letouvet B, Arvieux C, Leroy H, Polard J-L, Chaplain JM, Common H, et al. Predictors of failure for prosthetic joint infections treated with debridement. *Med Mal Infect.* 2016;46:39–43. doi:10.1016/j.medmal.2015.11.007.
- [43] Soriano A, García S, Bori G, Almela M, Gallart X, Maculé F, et al. Treatment of acute post-surgical infection of joint arthroplasty. *Clin Microbiol Infect.* 2006;12:930–933. doi:10.1111/j.1469-0691.2006.01463.x.
- [44] Kheir MM, Tan TL, Higuera C, George J, Della Valle CJ, Shen M, et al. Periprosthetic joint infections caused by Enterococci have poor outcomes. *J Arthroplasty.* 2017;32:933–947. doi:10.1016/j.arth.2016.09.017.
- [45] Tornero E, Senneville E, Euba G, Petersdorf S, Rodríguez-Pardo D, Lakatos B, et al. Characteristics of prosthetic joint infections due to *Enterococcus* sp. and predictors of failure: a multi-national study. *Clin Microbiol Infect.* 2014;20:1219–1224. doi:10.1111/1469-0691.12721.
- [46] Duijff SV, Vos FJ, Meis JF, Goosen JH. Debridement, antibiotics and implant retention in early postoperative infection with *Enterococcus* sp. *Clin Microbiol Infect.* 2015;21:e41–42. doi:10.1016/j.cmi.2015.01.006.

Authors: Erik Hansen, Jay Shah

QUESTION 10: Does performing a debridement, antibiotics and implant retention (DAIR) affect the outcome of a subsequent two-stage exchange arthroplasty?

RECOMMENDATION: Unknown. Based on the available evidence, it is not known if prior DAIR adversely affects the outcome of a subsequent two-stage exchange arthroplasty.

LEVEL OF EVIDENCE: Limited

DELEGATE VOTE: Agree: 93%, Disagree: 6%, Abstain: 1% (Super Majority, Strong Consensus)

RATIONALE

There are several surgical treatment options for periprosthetic joint infection (PJI), including irrigation and debridement (I&D) with modular component exchange and one- or two-stage exchange arthroplasty, with the ultimate choice depending on a number of variables, including chronicity of infection, organism and antibiotic sensitivity patterns, host factors and experience of surgeon. I&D with implant retention has been an attractive strategy in select circumstances as it is less morbid for the patient and less costly to the healthcare system overall. However, the failure rate of I&D is not insignificant, averaging 68% in the literature (61–82%). Following

treatment failure of an I&D, the recommendation for subsequent treatment is often a two-stage exchange arthroplasty. The question remains whether the initial attempt at I&D adversely affects the outcome of the subsequent two-stage exchange arthroplasty.

Two earlier studies and one very recent study on this subject seemed to indicate that failure of an initial I&D and modular component exchange leads to a higher than expected failure rates of subsequent two-stage exchange arthroplasty. Sherrell et al. performed a multicenter retrospective review of periprosthetic knee infections treated with a two-stage procedure following an initial treatment

with I&D [1]. Of the 83 knees that had undergone prior I&D, 28 (34%) failed subsequent two-stage revision and required reoperation for persistent infection. With the numbers available, there was no difference between success and failure with respect to age, gender or American Society of Anesthesiologists (ASA) grade. The other earlier study was a retrospective review of 44 patients who had undergone I&D for acute periprosthetic knee infections identified from the HealthEast Joint Replacement Registry and the Minneapolis Veterans Affairs Medical Center (MVAMC) total knee arthroplasty (TKA) database [2]. Of the 25 (57%) patients who failed an attempt at an I&D, 19 went on to an attempted two-stage revision procedure, and in only 11 of these 19 cases (58%) was the two-stage revision procedure ultimately successful. In a very recent retrospective review of 184 PJIs, Rajgopal et al. reported a 23.86% (21/88) failure rate after two-stage exchange following failed I&D compared to 15.62% (15/96) after direct two-stage exchange [3]. The success rate of the subsequent two-stage exchange arthroplasty procedures in all of these series is lower than historical published results, which the authors conclude may be due to the infection becoming more entrenched in the soft tissues and bone.

Two more recent studies on this topic report the opposite findings, namely that I&D before a two-stage exchange does not increase the risk of failure. Brimmo et al. used the California and New York State Inpatient Databases to identify all two-stage exchange revision TKA patients and compared failure rates, as defined as subsequent surgery due to infection within four years, between those with and without prior I&D [4]. Of the 750 patients who underwent two-stage exchange arthroplasty from 2005-2011, 57 (7.6%) had undergone a prior I&D. After four years, the estimated failure rate was 8.7% (95% confidence interval (CI), 1.9%-16.9%) in the group with prior I&D and 17.5% (95%CI, 14.7%-20.4%) in the group without prior I&D. After adjusting for sex, race, insurance, median household income and comorbidities, the hazard ratio for the group with a failed I&D was 0.49 ($p = 0.122$, 95% CI, 0.20-1.20) which the authors indicate revealed a lower risk of failure compared to the group without prior I&D. Nodzo

et al. reviewed their single institutional experience of patients who underwent two-stage exchange arthroplasty for PJI of total knee replacements, which included 132 who had not had an I&D and 45 patients who had a prior failed I&D [5]. The success rates between groups were similar at 82.5% and 82.2%, respectively, and the only variable they studied which decreased the odds of reoperation was the use of greater than 2gm of vancomycin in the spacer construct.

As is evident from the current literature, there is no conclusive evidence whether performing a DAIR affects the outcome of a subsequent two-stage exchange arthroplasty. All of the articles included, whether single institution, multicenter or database derived reported on a small number of patients who actually had a two-stage exchange arthroplasty after a failed I&D ($n = 83, 25, 88, 57, 45$) and therefore small differences in accuracy of coding or interpretation of data could potentially sway the results significantly. For those that support the belief that a failed I&D is associated with a decreased success rate for subsequent two-stage exchange arthroplasty, it may not be due to the infection becoming more established in the periarticular tissue, but that it is a patient or organism selection bias/confounding variable, and those individuals that fail an I&D inherently have a higher risk of failing a subsequent two-stage exchange arthroplasty.

REFERENCES

- [1] Sherrell JC, et al. The Chitranjan Ranawat Award: fate of two-stage reimplantation after failed irrigation and debridement for periprosthetic knee infection. *Clin Orthop Relat Res.* 2011;469:8-25.
- [2] Gardner J, Gioe TJ, Tatman P. Can this prosthesis be saved? Implant salvage attempts in infected primary TKA. *Clin Orthop Relat Res.* 2011;469:970-976.
- [3] Rajgopal A, et al. Does prior failed debridement compromise the outcome of subsequent two-stage revision done for periprosthetic joint infection following total knee arthroplasty? *J Arthroplasty.* 2018.
- [4] Brimmo O, et al. Irrigation and debridement before a 2-stage revision total knee arthroplasty does not increase risk of failure. *J Arthroplasty.* 2016;31:461-464.
- [5] Nodzo SR, et al. The influence of a failed irrigation and debridement on the outcomes of a subsequent 2-stage revision knee arthroplasty. *J Arthroplasty.* 2017;32:2508-2512.



Authors: Fabio Catani, Lazaros Poultsides, Henry Flores, Andrea Giorgini, Georgios K. Triantafyllopoulos, Arjun Saxena

QUESTION 11: How many debridement, antibiotics and implant retention (DAIR) procedure(s) are acceptable in management of patients with acute periprosthetic joint infection (PJI) of a primary arthroplasty before removal of components needs to be performed?

RECOMMENDATION: After one failed DAIR procedure, strong consideration should be given to removal of components.

LEVEL OF EVIDENCE: Limited

DELEGATE VOTE: Agree: 86%, Disagree: 13%, Abstain: 1% (Super Majority, Strong Consensus)

RATIONALE

A systematic review of the literature was conducted utilizing the Medline/PubMed (www.ncbi.nlm.nih.gov/pubmed), Embase (www.embase.com), and SCOPUS (www.scopus.com) databases. Studies in which there was a standard protocol for a second surgery other than DAIR (i.e., repeat surgery to remove antibiotic beads or planned multiple irrigation and debridement) were not included in this review.

The majority of the studies reviewed are limited by their retrospective nature, small sample sizes and lack of differentiation between acute postoperative PJI and late-hematogenous PJI. Most researchers viewed failure of DAIR as an indication for a different

therapeutic procedure; thus, most studies were limited to a single DAIR. Studies in which multiple DAIRs were performed had given limited insight in their methodology as to why and when a second procedure was performed. Multiple DAIR procedures were only performed in a small portion of the sample size [1,2].

A retrospective review by Triantafyllopoulos et al. attempted to address the appropriate number of DAIR procedures a patient should undergo before resection arthroplasty should be performed. In this retrospective series of 141 patients who underwent DAIR for treatment of a deep periprosthetic infection after primary or revision total knee arthroplasty (TKA) or total hip arthroplasty (THA),

19 patients underwent multiple DAIR procedures [3]. Of the 19 patients who underwent multiple (two or three) DAIR procedures, 10 (52.6%) achieved implant retention with infection control. Of the 122 patients who underwent a single DAIR, 78 (63.9%) achieved implant retention with infection control. All failures underwent prosthesis removal and two-stage reimplantation. The difference in failure rate between those who underwent multiple DAIR and those who underwent a single DAIR was not statistically significant. This study was limited by several factors. The authors included both primary and revision surgeries, as well as a heterogeneous mixture of acute postoperative PJI and late-hematogenous PJI. The manuscript also had no clear protocol for which patients underwent repeat DAIR or a different procedure. Furthermore, there was no protocol for patients to undergo additional DAIR or any notation of the timing. Patients who underwent a second DAIR greater than 20 days after the first DAIR had 97.4% lower odds of achieving success compared to patients undergoing the second procedure less than 20 days after the first [3].

A multicenter retrospective analysis by Urish et al. demonstrated 109 out of 216 patients who underwent DAIR after TKA required an additional procedure [4]. Of the 109 failures, 59 underwent repeat DAIR. Ultimately, of the patients who failed initial DAIR, only 28.4% had DAIR as their final procedure; thus, subsequent irrigation and debridement had a failure rate of over 70%.

Another retrospective study compared 64 patients who underwent DAIR ($n = 39$) versus two-stage revision ($n = 25$) within three months of primary TKA. Of the 39 patients who underwent DAIR, there were 24 failures (61.5%) and all 24 underwent repeat DAIR [5]. All 24 DAIR procedures failed to control the infection [5]. The DAIR patients underwent on average 3.2 additional surgical procedures

(range 1-6) to control the infection whereas the two-stage exchange patients underwent a mean of 2.2 surgical procedures (range 2-4). A further study by Vilchez et al. of 53 THA and TKA patients with PJI treated with DAIR, demonstrated that the need for a secondary DAIR was predictive of failure [6].

The literature demonstrates a second DAIR procedure has, at best, equivalent success as an initial DAIR procedure. In order to avoid additional surgical procedures, resection arthroplasty should be considered after an initial DAIR procedure.

REFERENCES

- [1] Mont MA, Waldman B, Banerjee C, Pacheco IH, Hungerford DS. Multiple irrigation, debridement, and retention of components in infected total knee arthroplasty. *J Arthroplasty*. 1997;12:426-433.
- [2] Azzam KA, Seeley M, Ghanem E, Austin MS, Purtill JJ, Parvizi J. Irrigation and debridement in the management of prosthetic joint infection: traditional indications revisited. *J Arthroplasty*. 2010;25:1022-1027. doi:10.1016/j.arth.2010.01.104.
- [3] Triantafyllopoulos G, Poultsides LA, Zhang W, Sculco PK, Ma Y, Sculco TP. Multiple irrigation and debridements for periprosthetic joint infections: facing a necessity or just prolonging the inevitable? *J Arthroplasty*. 2016;31:219-224. doi:10.1016/j.arth.2015.06.051.
- [4] Urish KL, Bullock AG, Kreger AM, Shah NB, Jeong K, Rothenberger SD, et al. A multicenter study of irrigation and debridement in total knee arthroplasty periprosthetic joint infection: treatment failure is high. *J Arthroplasty*. 2018;33:1154-1159. doi:10.1016/j.arth.2017.11.029.
- [5] Lizaur-Utrilla A, Gonzalez-Parreño S, Gil-Guillen V, Lopez-Prats FA. Debridement with prosthesis retention and antibiotherapy vs. two-stage revision for periprosthetic knee infection within 3 months after arthroplasty: a case-control study. *Clin Microbiol Infect*. 2015;21:851.e11-e17. doi:10.1016/j.cmi.2015.05.028.
- [6] Vilchez F, Martínez-Pastor JC, García-Ramiro S, Bori G, Maculé F, Sierra J, et al. Outcome and predictors of treatment failure in early post-surgical prosthetic joint infections due to *Staphylococcus aureus* treated with debridement. *Clin Microbiol Infect*. 2011;17:439-444. doi:10.1111/j.1469-0691.2010.03244.x.

Authors: Jamie Lora-Tamayo, David Warren, Mikel Mancheno-Losa, Marius Arndt, Christian Lausmann, Marius Arndt

QUESTION 12: What is the optimal length of antibiotic treatment following debridement, antibiotics and implant retention (DAIR) for acute periprosthetic joint infections (PJIs)?

RECOMMENDATION: The optimal length of antibiotic treatment following DAIR remains relatively unknown as there is considerable heterogeneity regarding the length, dose and administration of treatment. A minimum of six weeks of antibiotic therapy seems to be sufficient in most cases of PJIs managed by DAIR-provided surgical treatment.

LEVEL OF EVIDENCE: Moderate

DELEGATE VOTE: Agree: 91%, Disagree: 8%, Abstain: 1% (Super Majority, Strong Consensus)

RATIONALE

Acute PJIs may be treated by DAIR [1,2]. In this setting, antimicrobial therapy is administered at high doses during the postoperative period. The median success rate for DAIR for management of acute PJI varies from 34.8 - 100% [3-23]. However, none of the published reports directly compare the outcome of DAIR in relation to the length of antibiotic treatment.

In addition, the details of antibiotic treatment such as the route of administration, dose and the duration of therapy, appear to be missing. Two studies, though not providing the route of antimicrobial treatment, stated that patients undergoing DAIR in the cohort received at least six weeks and a median of seven weeks (range, 3 to 39 weeks) of antimicrobial treatment [9,10]. Majority of the studies reporting the outcome of DAIR [3,5,7,13-18] used an antibiotic treatment regimen based up the algorithm proposed

by Zimmerli et al. [1]. The latter consists of 7 to 14 days of intravenous antibiotics, followed by 3 to 6 months of oral antibiotics with activity against bacteria in biofilm (e.g., ciprofloxacin, adjunct therapy with rifampin).

Four studies report that intravenous antibiotic was used in their cohort, with or without adjunctive oral antibiotics during the course of treatment for a median duration of six weeks [8,12,19,24]. A single study discloses that the patients received oral antibiotics only after the DAIR procedure, with a duration of six weeks to life-long treatment [2]. The remaining 11 studies used a combination of intravenous, followed by oral antibiotic therapy. In these studies, the median duration of intravenous antibiotic therapy was 6 weeks and among the seven studies which reported the duration of oral antibiotics, the median was 16 weeks (range 9 weeks to lifelong).

TABLE 1.Comparative studies addressing the length of antimicrobial therapy in the setting of PJI managed by DAIR

Ref	Design	N	Etiology	Antimicrobials	Observations
26	Observational, retrospective, one center	112	Various	6 weeks of β -lactams/ glycopeptides, followed by oral treatment	Length of therapy did not predict the likelihood of failure
35	Observational, retrospective, comparative, non-randomized, one center	60	Various (mostly Staphylococci)	Common use of rifampin and ciprofloxacin	A 6-week treatment was non-inferior than a 12-week treatment
36	Observational, retrospective, comparative, pre-post design, one center	50	Various (mostly Staphylococci)	Common use of rifampin and fluoroquinolones	An 8-week treatment was non-inferior than long standard treatments (3-6 months)
37	Observational, retrospective, comparative, non-randomized, multicenter	87	Various (mostly Staphylococci)	Rifampin-based combinations	Same outcomes for 6-week and 12-week treatments
38	Multicenter Randomised Clinical Trial	63	Staphylococci	Levofloxacin + Rifampin	ITT analysis: 8-week treatment was non-inferior than 3-6 months. PP analysis : a trend towards non-inferiority was observed.

All studies included hip and knee prostheses. N, number of patients included (referring to those managed by debridement, antibiotics and implant retention); ITT, intention-to-treat; PP, per-protocol.

There appears to be a wide variation in the length of treatment, route of administration and the type of antimicrobial therapy that is selected for patients undergoing DAIR. The heterogeneity in the literature and the clinical practice may arise as a result of the fact that there are no reliable clinical or biological parameters that allows clinicians to assess the response to treatment and hence determine the optimal length of antimicrobial therapy [25]. There is a weak signal in the literature to suggest that after a "critical" period of antimicrobial therapy, no further improvement in outcome is encountered by extending the antimicrobial treatment. In fact, some investigators have stated that the length of antimicrobial therapy does not influence the outcome of treatment of PJI patients by DAIR [26]. To the contrary some investigators believe that prolonged antimicrobial therapy is more likely to lead to masking of the infection and a delay in identifying treatment failure [26,27].

There is little literature regarding the optimal route of administration of antimicrobial therapy. Majority of treating clinicians would recommend that patients undergoing DAIR should receive intravenous antimicrobials, at least initially. One observational non-randomized comparative study, concludes that the only factor associated with failure was the selection of oral antibiotics and not the duration of treatment [4]. The majority of studies that advocate the use of a six- to eight-week course of antibiotic therapy, state that intravenous antibiotics for two weeks followed by four to six weeks of oral antibiotics is optimal [27-34].

There are three observational non-randomized comparative studies showing no differences in success of DAIR when long or short course of antimicrobials were used (Table 1). In a study by Bernard et al., that included a cohort of 60 patients managed by DAIR, the success rate among patients treated for six weeks of antimicrobials was not lower than those treated for 12 weeks [35]. In 2012, Puhto et al. published a pre-post comparison of 50 patients with PJI treated for 8 weeks vs. 72 patients who received either 3 (hips) or 6 (knees) months of treatment, showing similar success rates (63 vs.67% in the intention-to-treat analysis, and 89 vs.87% in the per-protocol analysis) [36]. More recently, Chaussade et al. analyzed 87 episodes of PJI managed

by DAIR, with similar success rates when patients were treated for 6 or 12 weeks [37]. All three studies included knee and hip cases, all type of organisms with a predominance of Staphylococci and varying antibiotic regimen.

One randomized multicenter study compared an 8-week course of levofloxacin plus rifampin vs. a long course, three of oral therapy for hip PJI and six months of therapy for knee PJI in the setting of Staphylococcal PJI managed by DAIR [38]. Although the number of patients included was low, the non-inferiority hypothesis of the 8-week course was proven in the intention-to-treat analysis (success rate of 73 vs. 58% for the short course and long course groups, respectively; $n = 66$), and a trend towards non-inferiority was observed in the per-protocol analysis (cure rate of 92 and 95%; $n = 44$) [38]. The results of the DATIPO study, an ongoing French multicenter randomized clinical trial comparing 6 weeks vs. 12 weeks of antimicrobial therapy for patients with PJI undergoing surgical management, including DAIR, is eagerly awaited.

While the results of high level studies are awaited and based on the evaluation of the available literature, it appears that six to eight weeks of antimicrobial therapy is the ongoing standard for patients undergoing DAIR. There is less evidence regarding the optimal route of administration, with majority of the studies advocating the initial treatment should include intravenous route. The type of antimicrobials is also based on the organisms isolated with studies proposing that antibiotics targeting biofilm, such as rifampin, should also be part of the treatment algorithm.

REFERENCES

- [1] Zimmerli W, Ochsner PE. Management of infection associated with prosthetic joints. *Infection*. 2003;31:99-108. doi:10.1007/s15010-002-3079-9.
- [2] Kösters K, van Crevel R, Sturm PDJ, Willem Schreurs B, de Waal Malefijt MC, van Kampen A, et al. Treatment of knee prosthesis infections: evaluation of 15 patients over a 5-year period. *Int Orthop*. 2009;33:1249-1254. doi:10.1007/s00264-008-0638-4.
- [3] Leijtens B, Elbers JBW, Sturm PD, Kullberg BJ, Schreurs BW. Clindamycin-rifampin combination therapy for staphylococcal periprosthetic joint infections: a retrospective observational study. *BMC Infect Dis*. 2017;17:321. doi:10.1186/s12879-017-2429-2.

- [4] Tornero E, Morata L, Martínez-Pastor JC, Angulo S, Combalia A, Bori G, et al. Importance of selection and duration of antibiotic regimen in prosthetic joint infections treated with debridement and implant retention. *J Antimicrob Chemother.* 2016;71:1395–1401. doi:10.1093/jac/ckv481.
- [5] Senneville E, Joulie D, Legout L, Valette M, Dezeque H, Beltrand E, et al. Outcome and predictors of treatment failure in total hip/knee prosthetic joint infections due to *Staphylococcus aureus*. *Clin Infect Dis.* 2011;53:334–340. doi:10.1093/cid/cir402.
- [6] Peel TN, Buisson KL, Dowsey MM, Aboltins CA, Daffy JR, Stanley PA, et al. Outcome of debridement and retention in prosthetic joint infections by methicillin-resistant staphylococci, with special reference to rifampin and fusidic acid combination therapy. *Antimicrob Agents Chemother.* 2013;57:350–355. doi:10.1128/AAC.02061-12.
- [7] Berdal J-E, Skråmm I, Mowinckel P, Gulbrandsen P, Bjørnholt JV. Use of rifampicin and ciprofloxacin combination therapy after surgical debridement in the treatment of early manifestation prosthetic joint infections. *Clin Microbiol Infect.* 2005;11:843–845. doi:10.1111/j.1469-0691.2005.01230.x.
- [8] Mont MA, Waldman B, Banerjee C, Pacheco IH, Hungerford DS. Multiple irrigation, debridement, and retention of components in infected total knee arthroplasty. *J Arthroplasty.* 1997;12:426–433.
- [9] Westberg M, Grøgaard B, Snorrason F. Early prosthetic joint infections treated with debridement and implant retention: 38 primary hip arthroplasties prospectively recorded and followed for median 4 years. *Acta Orthop.* 2012;83:227–232. doi:10.3109/17453674.2012.678801.
- [10] Kuiper JWP, Vos SJC, Saouti R, Vergroesen DA, Graat HCA, Debets-Ossenkopp YJ, et al. Prosthetic joint-associated infections treated with DAIR (debridement, antibiotics, irrigation, and retention): analysis of risk factors and local antibiotic carriers in 91 patients. *Acta Orthop.* 2013;84:380–386. doi:10.3109/17453674.2013.823589.
- [11] Matsumoto T, Ishida K, Tsumura N, Nagai K, Muratsu H, Hida Y, et al. Treatment of 50 deep infections after total knee arthroplasty. *Orthopedics.* 2015;38:e529–e535. doi:10.3928/01477447-20150603-63.
- [12] Tsukayama DT, Estrada R, Gustilo RB. Infection after total hip arthroplasty. A study of the treatment of one hundred and six infections. *J Bone Joint Surg Am.* 1996;78:512–523.
- [13] Giulieri SG, Graber P, Ochsner PE, Zimmerli W. Management of infection associated with total hip arthroplasty according to a treatment algorithm. *Infection.* 2004;32:222–228. doi:10.1007/s15010-004-4020-1.
- [14] Kessler B, Knupp M, Graber P, Zwicky L, Hintermann B, Zimmerli W, et al. The treatment and outcome of peri-prosthetic infection of the ankle: a single cohort-centre experience of 34 cases. *Bone Joint J.* 2014;96-B:772–777. doi:10.1302/0301-620X.96B6.33298.
- [15] Letouvet B, Arvieux C, Leroy H, Polard JL, Chaplain JM, Common H, et al. Predictors of failure for prosthetic joint infections treated with debridement. *Med Mal Infect.* 2016;46:39–43. doi:10.1016/j.medmal.2015.11.007.
- [16] Sendi P, Löttscher PO, Kessler B, Graber P, Zimmerli W, Clauss M. Debridement and implant retention in the management of hip periprosthetic joint infection. *Bone Joint J.* 2017;99-B:330–336. doi:10.1302/0301-620X.99B3.BJF-2016-0609.R1.
- [17] Spormann C, Achermann Y, Simmen BR, Schwyzer H-K, Vogt M, Goldhahn J, et al. Treatment strategies for periprosthetic infections after primary elbow arthroplasty. *J Shoulder Elbow Surg.* 2012;21:992–1000. doi:10.1016/j.jse.2011.10.007.
- [18] Weenders SGM, Nijhof MW, Schimmel JJP, Goosen JHM. Debridement, antibiotics and implant retention in early periprosthetic joint infection after primary total hip arthroplasty: 88 percent survival after two years follow-up. *Acta Orthop Belg.* 2016;82:530–538.
- [19] Choi H-R, von Knoch F, Kandil AO, Zurakowski D, Moore S, Malchau H. Retention treatment after periprosthetic total hip arthroplasty infection. *Int Orthop.* 2012;36:723–729. doi:10.1007/s00264-011-1324-5.
- [20] Corona Pérez-Cardona PS, Barro Ojeda V, Rodríguez Pardo D, Pigrau Serrallach C, Guerra Farfán E, Amat Mateu C, et al. Clinical experience with daptomycin for the treatment of patients with knee and hip periprosthetic joint infections. *J Antimicrob Chemother.* 2012;67:1749–1754. doi:10.1093/jac/dks119.
- [21] Zhang C, Yan CH, Chan PK, Ng FY, Chiu KY. Polyethylene insert exchange is crucial in debridement for acute periprosthetic infections following total knee arthroplasty. *J Knee Surg.* 2017;30:36–41.
- [22] Dennison T, Alentorn-Geli E, Assenmacher AT, Sperling JW, Sánchez-Sotelo J, Cofield RH. Management of acute or late hematogenous infection after shoulder arthroplasty with irrigation, debridement, and component retention. *J Shoulder Elbow Surg.* 2017;26:73–78. doi:10.1016/j.jse.2016.05.018.
- [23] Hyman JL, Salvati EA, Laurencin CT, Rogers DE, Maynard M, Brause DB. The arthroscopic drainage, irrigation, and debridement of late, acute total hip arthroplasty infections: average 6-year follow-up. *J Arthroplasty.* 1999;14:903–910.
- [24] Osmon DR, Berbari EF, Berendt AR, Lew D, Zimmerli W, Steckelberg JM, et al. Diagnosis and management of prosthetic joint infection: clinical practice guidelines by the Infectious Diseases Society of America. *Clin Infect Dis.* 2013;56:e1–e25. doi:10.1093/cid/cis803.
- [25] Pisto RJ, Elke R. Antibiotic treatment can be safely stopped in asymptomatic patients with prosthetic joint infections despite persistent elevated C-reactive protein values. *Infection.* 2010;38:293–296. doi:10.1007/s15010-010-0019-y.
- [26] Byren I, Bejon P, Atkins BL, Angus B, Masters S, McLardy-Smith P, et al. One hundred and twelve infected arthroplasties treated with “DAIR” (debridement, antibiotics and implant retention): antibiotic duration and outcome. *J Antimicrob Chemother.* 2009;63:1264–1271. doi:10.1093/jac/dkp107.
- [27] Fink B, Schuster P, Schwenninger C, Frommelt L, Oremek D. A standardized regimen for the treatment of acute postoperative infections and acute hematogenous infections associated with hip and knee arthroplasties. *J Arthroplasty.* 2017;32:1255–1261. doi:10.1016/j.arth.2016.10.011.
- [28] Triantafyllopoulos GK, Soranoglou V, Memtsoudis SG, Poulosides LA. Implant retention after acute and hematogenous periprosthetic hip and knee infections: whom, when and how? *World J Orthop.* 2016;7:546–552. doi:10.5312/wjo.v7.i9.546.
- [29] Lee HD, Prashant K, Shon WY. Management of periprosthetic hip joint infection. *HP J.* 2015;27:63–71. doi:10.5371/hp.2015.27.2.63.
- [30] Van Kleunen JP, Knox D, Garino JP, Lee GC. Irrigation and debridement and prosthesis retention for treating acute periprosthetic infections. *Clin Orthop Relat Res.* 2010;468:2024–2028. doi:10.1007/s11999-010-1291-y.
- [31] Koyonos L, Zmisteowski B, Della Valle CJ, Parvizi J. Infection control rate of irrigation and debridement for periprosthetic joint infection. *Clin Orthop Relat Res.* 2011;469:3043–3048. doi:10.1007/s11999-011-1910-2.
- [32] Koh JJ, Han SB, In Y, Oh KJ, Lee DH, Kim TK, et al. Open debridement and prosthesis retention is a viable treatment option for acute periprosthetic joint infection after total knee arthroplasty. *Arch Orthop Trauma Surg.* 2015;135:847–855. doi:10.1007/s00402-015-2237-3.
- [33] Farhad R, Roger P-M, Albert C, Pelligri C, Touati C, Dellamonica P, et al. Six weeks antibiotic therapy for all bone infections: results of a cohort study. *Eur J Clin Microbiol Infect.* 2010;29:217–222. doi:10.1007/s10096-009-0842-1.
- [34] Armstrong MD, Carli AV, Abdelbary H, Poitras S, Lapner P, Beaulé PE. Tertiary care centre adherence to unified guidelines for management of periprosthetic joint infections: a gap analysis. *Can J Surg J Can Chir.* 2018;61:34–41.
- [35] Bernard L, Legout L, Zürcher-Pfund L, Stern R, Rohner P, Peter R, et al. Six weeks of antibiotic treatment is sufficient following surgery for septic arthroplasty. *J Infect.* 2010;61:125–132. doi:10.1016/j.jinf.2010.05.005.
- [36] Puhto A-P, Puhto T, Syrjala H. Short-course antibiotics for prosthetic joint infections treated with prosthesis retention. *Clin Microbiol Infect.* 2012;18:1143–1148. doi:10.1111/j.1469-0691.2011.03693.x.
- [37] Chaussade H, Uçkay I, Vuagnat A, Druon J, Gras G, Rosset P, et al. Antibiotic therapy duration for prosthetic joint infections treated by Debridement and Implant Retention (DAIR): Similar long-term remission for 6 weeks as compared to 12 weeks. *Int J Infect Dis.* 2017;63:37–42. doi:10.1016/j.ijid.2017.08.002.
- [38] Lora-Tamayo J, Euba G, Cobo J, Horcajada JP, Soriano A, Sandoval E, et al. Short- versus long-duration levofloxacin plus rifampicin for acute staphylococcal prosthetic joint infection managed with implant retention: a randomised clinical trial. *Int J Antimicrob Agents.* 2016;48:310–316. doi:10.1016/j.ijantimicag.2016.05.021.

Authors: Camelia Marculescu, Silvano Esposito

QUESTION 13: What is the most effective combination of antibiotics in the treatment of acute periprosthetic joint infections (PJIs) caused by methicillin-resistant *Staphylococcus aureus* (MRSA) that has undergone surgical management with debridement, antibiotics and implant retention (DAIR)?

RECOMMENDATION: We recommend a combination of a parenteral antibiotic plus oral rifampin for one to six weeks, followed by rifampin and a companion highly bioavailable oral drug for additional three months, depending on the susceptibility profile of MRSA, patient tolerability and side effect profile.

LEVEL OF EVIDENCE: Limited

DELEGATE VOTE: Agree: 88%, Disagree: 10%, Abstain: 2% (Super Majority, Strong Consensus)

RATIONALE

Treatment of MRSA PJI that has undergone DAIR remains challenging. An ideal combination of antimicrobial therapy has not been established. Treatment should take into account antimicrobial susceptibilities of MRSA and tailored accordingly. Whenever possible, rifampin-based combinations should be used, but rifampin alone should never be used due to the rapid development of resistance. Rifampin-based combination therapy regimens have been shown to be effective in eradication of staphylococcal organisms and cure PJIs. A widely used algorithm by Zimmerli and the Infectious Diseases Society of America (IDSA) guidelines recommend a quinolone–rifampin combination for susceptible *Staphylococcus aureus* strains and cure rates of 70–100% have been reported [1–3]. The duration of antimicrobial therapy for PJI managed with DAIR has not been well established. We recommend two to six weeks of parenteral antimicrobial therapy in combination with rifampin 300 to 450 mg orally twice a day, followed by rifampin plus a susceptible companion oral drug (such as trimethoprim-sulfamethoxazole, ciprofloxacin or levofloxacin, a tetracycline, fusidic acid) depending on the individual tolerance, side effect profile and antimicrobial susceptibility testing [1,4,5]. Certain highly bioavailable drugs such as fluoroquinolones, rifampin, linezolid and trimethoprim-sulfamethoxazole, reach levels in bone that exceed the minimal inhibitory concentration (MICs) for most organisms [6].

Zimmerli et al. have suggested a duration of therapy of three months for total hip arthroplasties (THAs) PJIs and six months for total knee arthroplasties (TKAs) PJIs [1,3]. Shorter courses of therapy (6 vs. 12 weeks) were studied in PJIs treated with DAIR. However, in this study by Chaussade et al. the presence of MRSA, which comprised only 13.8% of infections, was associated with a poorer outcome (remission in 41.7 vs. 73.3% for other pathogens [7]. Chronic oral suppression with trimethoprim-sulfamethoxazole, minocycline or doxycycline based on in vitro-susceptibilities and individual side effect profile and tolerance may be considered following the above regimens and should be reserved for patients who are unsuitable or refuse further surgical therapy. The duration of chronic oral suppression remains unknown.

While the current IDSA guidelines recommend vancomycin as the primary parenteral agent for treatment of MRSA infections, its utility has been questioned due to increasing reports of heterogeneous resistance, treatment failure, and nephrotoxicity. Vancomycin is not bactericidal against small colony variants (SCV) of MRSA. Moreover, Lenhard et al. showed recently in mixed-population experiments that vancomycin favorably selects for the growth of

the SCV subpopulation [6]. Therefore, clinicians should consider glycopeptide combination regimens or alternative antimicrobials in patients with severe persistent MRSA infections in which the SCV phenotype may play a role.

In vitro analyses have identified fluoroquinolones and oritavancin as retaining high levels of vancomycin in vitro against SCVs and β -lactam combinations with daptomycin may offer a new option for combating SCVs [8,9,10]. While optimal treatment for infections caused by staphylococcal SCVs is not known, combination therapy including either rifampin or oritavancin appears to be particularly effective at eradicating intracellular SCVs [11].

REFERENCES

- [1] Osmon DR, Berbari EF, Berendt AR, Lew D, Zimmerli W, Steckelberg JM, et al. Diagnosis and management of prosthetic joint infection: clinical practice guidelines by the Infectious Diseases Society of America. *Clin Infect Dis*. 2013;56:e1–e25. doi:10.1093/cid/cis803.
- [2] Trampuz A, Zimmerli W. Antimicrobial agents in orthopaedic surgery: Prophylaxis and treatment. *Drugs*. 2006;66:1089–1105.
- [3] Zimmerli W, Trampuz A, Ochsner PE. Prosthetic-joint infections. *N Engl J Med*. 2004;351:1645–1654. doi:10.1056/NEJMra040181.
- [4] Leijtens B, Elbers JBW, Sturm PD, Kullberg BJ, Schreurs BW. Clindamycin-rifampin combination therapy for staphylococcal periprosthetic joint infections: a retrospective observational study. *BMC Infect Dis*. 2017;17:321. doi:10.1186/s12879-017-2429-2.
- [5] Peel TN, Buising KL, Dowsey MM, Aboltins CA, Daffy JR, Stanley PA, et al. Outcome of debridement and retention in prosthetic joint infections by methicillin-resistant staphylococci, with special reference to rifampin and fusidic acid combination therapy. *Antimicrob Agents Chemother*. 2013;57:350–355. doi:10.1128/AAC.02061-12.
- [6] Lenhard JR, von Eiff C, Hong IS, Holden PN, Bear MD, Suen A, et al. Evolution of *Staphylococcus aureus* under vancomycin selective pressure: the role of the small-colony variant phenotype. *Antimicrob Agents Chemother*. 2015;59:1347–1351. doi:10.1128/AAC.04508-14.
- [7] Chaussade H, Uçkay I, Vuagnat A, Druon J, Gras G, Rosset P, et al. Antibiotic therapy duration for prosthetic joint infections treated by Debridement and Implant Retention (DAIR): Similar long-term remission for 6 weeks as compared to 12 weeks. *Int J Infect Dis*. 2017;63:37–42. doi:10.1016/j.ijid.2017.08.002.
- [8] Begic D, von Eiff C, Tsuji BT. Daptomycin pharmacodynamics against *Staphylococcus aureus* hemB mutants displaying the small colony variant phenotype. *J Antimicrob Chemother*. 2009;63:977–981. doi:10.1093/jac/dkp069.
- [9] Mehta S, Singh C, Plata KB, Chanda PK, Paul A, Riosa S, et al. β -Lactams increase the antibacterial activity of daptomycin against clinical methicillin-resistant *Staphylococcus aureus* strains and prevent selection of daptomycin-resistant derivatives. *Antimicrob Agents Chemother*. 2012;56:6192–6200. doi:10.1128/AAC.01525-12.
- [10] Ortwine JK, Werth BJ, Sakoulas G, Rybak MJ. Reduced glycopeptide and lipopeptide susceptibility in *Staphylococcus aureus* and the “seesaw effect”: Taking advantage of the back door left open? *Drug Resist Updat*. 2013;16:73–79. doi:10.1016/j.drug.2013.10.002.
- [11] Massey RC, Peacock SJ. Antibiotic-resistant sub-populations of the pathogenic bacterium *Staphylococcus aureus* confer population-wide resistance. *Curr Biol*. 2002;12:R686–R687.



Authors: Jean Yombi, Marjan Wouthuyzen-Bakker

QUESTION 14: What antibiotic therapy (agent, route, dose and duration) is recommended for gram-negative acute periprosthetic joint infections (PJIs) being treated with debridement, antibiotics and implant retention (DAIR)?

RECOMMENDATION: Following surgical intervention (DAIR), gram-negative acute PJI patients should also receive antibiotic treatment for 6 to 12 weeks based on the type of organism. In fluoroquinolone-susceptible cases, the recommended antibiotic agent is a fluoroquinolone.

LEVEL OF EVIDENCE: Moderate

DELEGATE VOTE: Agree: 83%, Disagree: 11%, Abstain: 6% (Super Majority, Strong Consensus)

RATIONALE

In recent decades, the number of PJIs caused by gram-negative organisms, including multidrug-resistant gram-negatives (GNs), has increased [1]. Several studies have been published on antibiotic treatment of these infections in patients treated with surgical debridement and implant retention (DAIR) [2–8]. Studies have been performed demonstrating the preferred antibiotic agent for treating these infections, but few relate to the preferred route, dose and duration of antibiotic treatment.

Antibiotic Agent for GN PJIs Treated with DAIR

Rodriguez-Pardo et al. performed a retrospective analysis on 242 GN PJIs, including 174 cases (72%) treated with DAIR [2]. The study demonstrated that the use of fluoroquinolones (in this study ciprofloxacin) was associated with the highest success rate of 79% (98 of 124), while the success in the remainder of the patients treated with other antibiotic regimen (e.g., β -lactam or cotrimoxazole) was only 40% (20 of 49). In addition, ciprofloxacin treatment exhibited an independent protective effect in the prevention of subsequent failure in the multivariate analysis (adjusted hazard ratio (aHR) 0.23; $p < 0.001$). In addition to endorsing the use of fluoroquinolones, the latter study also favored the use of combination therapy, as a β -lactam antibiotic combined with a fluoroquinolone or an aminoglycoside as this regimen showed a trend towards better outcome (aHR 0.42, $p < 0.07$). The cohort of patients included in the study were mostly infected with *Enterobacteriaceae* spp. (78%) and some with *Pseudomonas* spp. (20%). The study was not able to glean which of the PJI cases benefited from the combination therapy. Several other smaller studies have been performed, supporting the beneficial effect of fluoroquinolones. Aboltins et al. [3] studied the outcome of 17 consecutive patients with an early GN PJI, mostly polymicrobial in origin (76%), and mainly involving *Enterobacteriaceae* spp (94%). All of these patients were initially treated with β -lactam antibiotics intravenously, and 14 patients were subsequently treated with oral ciprofloxacin. Treatment failure occurred in two patients not treated with ciprofloxacin (median period of follow-up of 28 months). Only one of these failures was caused by a relapse with the same GN, suggesting a cure rate of 100% (14/14) when using ciprofloxacin versus 66% (2/3) when using another oral antibiotic regimen (in these particular cases amoxicillin/clavulanic acid). In addition, a study

performed by Jaén et al. ($n = 47$) and Tornero et al. ($n = 21$) on GN PJIs treated with DAIR, which were partly based on the same cohort of patients, also demonstrated that the use of fluoroquinolones in susceptible GN was the only factor associated with treatment success in the univariate analysis [4,7,8].

Recently, Grossi et al. [9] demonstrated in 76 GN PJIs that the outcome of treatment with IV β -lactam antibiotics (alone or in combination with another antimicrobial agent) during the whole treatment period (median three months) was similar compared to the use of an oral fluoroquinolone (failure rate 16.7 vs. 22.4%, $p = 0.75$). Although the study of Grossi et al. included both DAIRs and revisions as surgical strategy, outcome remained the same after stratification according to the surgical procedure, suggesting that intravenously antibiotic regimens and/or combination therapy may be as effective as treatment with fluoroquinolones.

The use of alternative oral regimens other than β -lactam, like cotrimoxazole, have been poorly studied in the field of PJI and require further investigation.

Only a few data are available on how to treat multidrug-resistant (MDR) GN in the field of PJIs, but extensive reviews and expert opinions have been published, utilizing the efficacy of carbapenems, combined with tigecycline, colistin or fosfomycin when the microorganism is susceptible [10–13]. Another question in the consensus document elaborates on the efficacy of tigecycline and fosfomycin alone or in conjunction with β -lactam in the treatment of PJI, suggesting that tigecycline or fosfomycin could be considered for the treatment of MDR GN PJI of as a part of a combination regimen when the microorganism is susceptible. In addition, the benefit of adding colistin to a β -lactam for osteoarticular infections caused by MDR, have been reported as well, demonstrating a higher cure rate for combination therapy [14,15].

Treatment Duration, Route and Dosage for GN PJIs Treated with DAIR

Table 1 shows the treatment duration and subsequent failure rate of the above-mentioned studies. Whether a short or long treatment duration was associated with a respectively lower or higher cure rate was not described in most studies. Only Jaén et al. evaluated the difference in outcome between patients treated with more or less

TABLE 1. Overview treatment duration and outcome in GN PJIs solely treated with DAIR

Author, Year	Patients (n)	IV (days)	Oral (days)	Total (days)	Failure %
Tornero et al. 2016 [4]	21	8 (IQR 5-12) [#]	69 (IQR 45-95) [#]	ND	14
Grossi et al. 2016 [9]	35	36 (IQR 14-90) [*]	ND	90 (IQR 89-92) [*]	23
Jaén et al. 2012 [8]	47	14 (IQR 8-24)	64 (IQR 28-102)	ND	26
Rodriguez-Pardo et al. 2011 [2]	174	14 (IQR 6-23)	58 (IQR 27-90).	ND	32
Zmistowski et al. 2011 [5]	10	ND	ND	ND	30
Aboltins et al. 2011 [3]	17	40 (range, 9 - 79)	365 (range, 30 - 1678).	ND	6
Hsieh et al. 2009 [6]	27	38 (range, 24-52)	49 (range, 28-92)	ND	27

^{*}, duration of treatment included cases treated with revision surgery; [#], duration of treatment included gram-positive PJIs; IQR, interquartile range; ND, no data.

than 14 days of IV treatment and treated with more or less than 64 days of oral antibiotic treatment and demonstrated no differences in outcome [8]. Although studies have demonstrated an equal success rate with 6 to 8 weeks compared to the standard 12 weeks of antibiotic treatment [16–20], these studies have been mainly performed in rifampin susceptible staphylococci and cannot be extrapolated to GN PJI. For this reason, we would still recommend a 6 to 12-week treatment duration (including 1 to 2 weeks of IV treatment), especially in ciprofloxacin-resistant GN. In case β -lactam is indicated, it should be administered intravenously throughout the entire treatment period.

No studies evaluated the dosage of antibiotic treatment and its relation to outcome. We propose the recommendations depicted in Table 2.

REFERENCES

- [1] Benito N, Franco M, Ribera A, Soriano A, Rodríguez-Pardo D, Sorli L, et al. Time trends in the aetiology of prosthetic joint infections: a multicentre cohort study. *Clin Microbiol Infect.* 2016;22:732.e1–e8. doi:10.1016/j.cmi.2016.05.004.
- [2] Rodríguez-Pardo D, Pigrau C, Lora-Tamayo J, Soriano A, del Toro MD, Cobo J, et al. Gram-negative prosthetic joint infection: outcome of a debridement, antibiotics and implant retention approach. A large multicentre study. *Clin Microbiol Infect.* 2014;20:O911–O919. doi:10.1111/1469-0691.12649.
- [3] Aboltins CA, Dowsey MM, Buisson KL, Peel TN, Daffy JR, Choong PFM, et al. Gram-negative prosthetic joint infection treated with debridement, prosthesis retention and antibiotic regimens including a fluoroquinolone. *Clin Microbiol Infect.* 2011;17:862–867. doi:10.1111/j.1469-0691.2010.03361.x.
- [4] Tornero E, Morata L, Martínez-Pastor JC, Angulo S, Combalia A, Bori G, et al. Importance of selection and duration of antibiotic regimen in prosthetic joint infections treated with debridement and implant retention. *J Antimicrob Chemother.* 2016;71:1395–1401. doi:10.1093/jac/dkv481.
- [5] Zmistowski B, Fedorka CJ, Sheehan E, Deirmengian G, Austin MS, Parvizi J. Prosthetic joint infection caused by gram-negative organisms. *J Arthroplasty.* 2011;26:104–8. doi:10.1016/j.arth.2011.03.044.
- [6] Hsieh PH, Lee MS, Hsu KY, Chang YH, Shih HN, Ueng SW. Gram-negative prosthetic joint infections: risk factors and outcome of treatment. *Clin Infect Dis.* 2009;49:1036–1043. doi:10.1086/605593.
- [7] Martínez-Pastor JC, Muñoz-Mahamud E, Vilchez F, García-Ramiro S, Bori G, Sierra J, et al. Outcome of acute prosthetic joint infections due to gram-negative bacilli treated with open debridement and retention of the prosthesis. *Antimicrob Agents Chemother.* 2009;53:4772–4777. doi:10.1128/AAC.00188-09.
- [8] Jaén N, Martínez-Pastor JC, Muñoz-Mahamud E, García-Ramiro S, Bosch J, Mensa J, et al. Long-term outcome of acute prosthetic joint infections due to gram-negative bacilli treated with retention of prosthesis. *Rev Esp Quimioter.* 2012;2:194–198.
- [9] Grossi O, Asseray N, Bourigault C, Corvec S, Valette M, Navas D, et al. Gram-negative prosthetic joint infections managed according to a multidisciplinary standardized approach: risk factors for failure and outcome with and without fluoroquinolones. *J Antimicrob Chemother.* 2016;71:2593–2597. doi:10.1093/jac/dkw202.
- [10] Perez-Jorge C, Gomez-Barrena E, Horcajada J-P, Puig-Verdie L, Esteban J. Drug treatments for prosthetic joint infections in the era of multidrug resistance. *Expert Opin Pharmacother.* 2016;17:1233–1246. doi:10.1080/14656566.2016.1176142.

TABLE 2. Proposed antibiotic regimen for GN PJIs treated with DAIR

Microorganisms ¹	IV Regimen	Oral Regimen
<i>Enterobacteriaceae</i> , ciprofloxacin susceptible	Ceftriaxon 2 gm QD ± Ciprofloxacin 400 mg TID	Ciprofloxacin 750 mg BID
<i>Pseudomonas</i> spp, ciprofloxacin susceptible	Cefepime 2 gm TID <i>or</i> Meropenem 2gm TID <i>or</i> Ceftazidime 2gm TID <i>or</i> Piperacillin-tazobactam 4.5gr QID ± Ciprofloxacin 400 mg TID <i>or</i> Tobramycin 7mg/kg QD	Ciprofloxacin 750 mg BID
<i>Enterobacteriaceae</i> , ciprofloxacin-resistant	Ceftriaxone 2 gm QD ± Tobramycin 7mg/kg QD	IV β -lactam antibiotics during the whole treatment period <i>Possible alternative</i> Cotrimoxazole 960 mg TID
<i>Pseudomonas</i> spp, ciprofloxacin resistant	Cefepime 2 gm TID <i>or</i> Meropenem 2gm TID <i>or</i> Ceftazidime 2gm TID <i>or</i> Piperacillin-tazobactam 4.5gr QID ± Tobramycin 7mg/kg QD <i>or</i> Colistin 3 million IU TID <i>or</i> Fosfomycin 2-4g QID	IV antibiotics during the whole treatment period

DAIR, debridement, antibiotics and implant retention; PJIs, periprosthetic joint infections; QD, four times daily; TID, three times daily; BID, twice daily

± Duo therapy can be considered in patients who have a high risk for treatment failure.

¹ In case of multidrug-resistant or extremely drug-resistant gram-negative, the antibiotic treatment should be guided by the antibiogram and preferentially by combining two antibiotics with a different mechanism of action.

- [11] de Sanctis J, Teixeira L, van Duin D, Odio C, Hall G, Tomford JW, et al. Complex prosthetic joint infections due to carbapenemase-producing *Klebsiella pneumoniae*: a unique challenge in the era of untreatable infections. *Int J Infect Dis*. 2014;25:73–78. doi:10.1016/j.ijid.2014.01.028.
- [12] Rodríguez-Baño J, Gutiérrez-Gutiérrez B, Machuca I, Pascual A. Treatment of infections caused by extended-spectrum-beta-lactamase-, AmpC-, and carbapenemase-producing enterobacteriaceae. *Clin Microbiol Rev*. 2018;31. doi:10.1128/CMR.00079-17.
- [13] Tumbarello M, Viale P, Bassetti M, De Rosa FG, Spanu T, Viscoli C. Infections caused by KPC-producing *Klebsiella pneumoniae*: differences in therapy and mortality in a multicentre study—authors' response. *J Antimicrob Chemother*. 2015;70:2922. doi:10.1093/jac/dkv200.
- [14] Ribera A, Benavent E, Lora-Tamayo J, Tubau F, Pedrero S, Cabo X, et al. Osteoarticular infection caused by MDR *Pseudomonas aeruginosa*: the benefits of combination therapy with colistin plus β -lactams. *J Antimicrob Chemother*. 2015;70:3357–3365. doi:10.1093/jac/dkv281.
- [15] Lora-Tamayo J, Murillo O, Bergen PJ, Nation RL, Poudyal A, Luo X, et al. Activity of colistin combined with doripenem at clinically relevant concentrations against multidrug-resistant *Pseudomonas aeruginosa* in an in vitro dynamic biofilm model. *J Antimicrob Chemother*. 2014;69:2434–2442. doi:10.1093/jac/dku151.
- [16] Chaussade H, Uçkay I, Vuagnat A, Druon J, Gras G, Rosset P, et al. Antibiotic therapy duration for prosthetic joint infections treated by Debridement and Implant Retention (DAIR): Similar long-term remission for 6 weeks as compared to 12 weeks. *Int J Infect Dis*. 2017;63:37–42. doi:10.1016/j.ijid.2017.08.002.
- [17] Cunningham DJ, Kavolus JJ, Bolognesi MP, Wellman SS, Seyler TM. Specific infectious organisms associated with poor outcomes in treatment for hip periprosthetic infection. *J Arthroplasty*. 2017;32:1984–1990.e5. doi:10.1016/j.arth.2017.01.027.
- [18] Bernard L, Legout L, Zürcher-Pfund L, Stern R, Rohner P, Peter R, et al. Six weeks of antibiotic treatment is sufficient following surgery for septic arthroplasty. *J Infect*. 2010;61:125–132. doi:10.1016/j.jinf.2010.05.005.
- [19] Farhad R, Roger P-M, Albert C, Pelligri C, Touati C, Dellamonica P, et al. Six weeks antibiotic therapy for all bone infections: results of a cohort study. *Eur J Clin Microbiol Infect*. 2010;29:217–222. doi:10.1007/s10096-009-0842-1.
- [20] Puhto A-P, Puhto T, Syrjala H. Short-course antibiotics for prosthetic joint infections treated with prosthesis retention. *Clin Microbiol Infect*. 2012;18:1143–1148. doi:10.1111/j.1469-0691.2011.03693.x.

5.3. TREATMENT: ONE-STAGE EXCHANGE

Authors: Navin Fernando, Pedro Foguet, Michael A. Mont, Nipun Sodhi, Robert Molloy, Ariel Saldaña

QUESTION 1: What are the potential advantages of a one-stage exchange arthroplasty?

RECOMMENDATION: The potential advantages of a one-stage exchange arthroplasty are multiple, including a decrease in surgical morbidity and mortality, earlier functional return, decrease in healthcare and global economic costs as well as an increase in health-related quality adjusted life years.

LEVEL OF EVIDENCE: Moderate

DELEGATE VOTE: Agree: 89%, Disagree: 8%, Abstain: 3% (Super Majority, Strong Consensus)

RATIONALE

While multiple studies have been performed evaluating the efficacy of a one-stage or two-stage exchange arthroplasty for periprosthetic joint infection (PJI) [1–13], the majority demonstrated a reduced rate of recurrent infection after a two-stage exchange in comparison to a one-stage exchange, although the comparative value of these results is difficult to interpret given discrepancies in patient comorbidities, bacterial profiles, treatment protocols as well as variances in the definitions of PJIs, clinical success, and failure.

In North America, treatment of PJIs using a two-stage revision procedure remains the most widely utilized and reported method in the literature [14–16]. However, there is no clear evidence that shows superiority of two-stage over one-stage revision in terms of success, eradication of infection or patient satisfaction [1–11,13,16–18]. In addition, one-stage revision has demonstrated multiple advantages in several prognostic and observational studies, particularly within the European literature [1–13].

Depending on the study and follow-up time, one-stage revision procedures have demonstrated a success rate ranging between 75 to 95% [1–5,7–13,17–19]. This is comparable to the reported reinfection rates after two-stage revisions between 9 and 20% of cases [20]. Furthermore, when appropriately performed, one-stage revision can avoid the morbidity associated with multiple surgeries while providing the advantages of reduced total length of stay, overall cost and earlier functional rehabilitation [19,20]. Other advantages include the reduced duration of postoperative systemic antibiotic therapy and systemic antibiotic side effects [19,20].

Despite this demonstrated success of one-stage revisions, it is critical to recognize that this procedure is contingent on strict

patient selection criteria and specific operative planning protocols. For example, preoperative identification of the responsible bacterial organism in the synovial fluid is a prerequisite to determine the specific local and systemic antibiotic therapy regimen [3,6,10,11,19]. Also, patients who fail prior one-stage revision, those with an unclear causative pathogen or lack of susceptibility to available antibiotics and those with more extensive infections, may not be candidates for one-stage exchange [20].

In addition to strict selection criteria, several meticulous intraoperative steps, including aggressive soft tissue debridement, meticulous removal of the prior cement material and all hardware, as well as the use of antibiotic-loaded cement for reimplantation, along with specific postoperative antibiotic regimens, are important for success [19]. In a systematic review comparing one- to two-stage exchange, superior outcomes for one-stage revision were reported when performed in this selective patient population [21].

Two recent meta-analyses comparing outcomes for one-stage versus two-stage exchange for patients who have PJIs after both total hip [22] and total knee [23] arthroplasties demonstrated statistically equivalent reinfection rates for both protocols. These findings, were, however limited by the quality of the studies included in the meta-analyses, as well as a relative paucity of studies evaluating one-stage protocols in comparison to two-stage exchange.

Wolf et al. utilized Markov modeling in a decision-tree analysis to suggest a possible superiority of treatment of a one-stage exchange in comparison to a two-stage protocol as it pertains to health-related quality of life years, despite an objective decrease in recurrent infection with a two-stage protocol [24]. Although